DEVS-PWSA: A Mission Engineering Engine Designed for Tranche Evolution

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(1) Glossary

AFSIM	Advanced Framework for Simulation, Integration, and Modeling
AI	Artificial Intelligence
BFO	Basic Formal Ontology
CAMLI	Collaborative Development Architecture for Model Level Integration
CONOPS	Concept of Operations
DE	Digital Engineering
DEF	Digital Engineering Factory
DEVS	Discrete Event Systems Specification
DoD	Department of Defense
DoW	Department of War
dTEMP	Digital Test and Evaluation Master Plan
dT&E	Digital Test & Evaluation
DVAST	DEVS Virtual As-Is Simulation Test Platform
EF	Experimental Frame
GAO	Government Accountability Office
IDSK	Integrated Decision Support Key
KPI	Key Performance Indicator
KPP	Key Performance Parameters
LEO	Low-Earth Orbit
LLM	Large Language Model
MBSE	Model-Based Systems Engineering
MDA	Missile Defense Agency
ME	Mission Engineering
MS&A	Modeling, Simulation, and Analysis
MoP	Measures of Performance
M&S	Modeling and Simulation
NTN	Non-Terrestrial Network
OCT	Optical Communication Terminals
OS	Ontology Stack
OWL	WebOntologyLanguage
PROV-O	Provenance Ontology
PT&E	Phased Test and Evaluation

PWSA	Proliferated Warfighter Space Architecture
RAG	Retrieval Augmented Generation
RDF	Resource Description Framework
RF	Radio Frequency
SDA	Space Development Agency
SE	System Engineering
SES	System Entity Structure
SHACL	Shapes Constraint Language
SLM	Small Language Model
SoS	System of Systems
STOIC	Systems Theoretic Ontological Integration Core
SysML	Systems Modeling Language
T3SD	The Tricotyledon Theory of System Design
TPOC	Technical Point Of Contact
T&E	Test & Evaluation
UA	University of Arizona
V&V	Verification and Validation
VV&A	Verification, Validation, and Accreditation

(2) Milestone Identification

Table 1: *Program Schedule with all Key Milestones*

Milestone	**Expected Delivery (Months after CA)**	**Deliverable**	**Acceptance Criteria**
Task 1 Kickoff: Formal Substrate Initialization	+2	Requirements analysis, ontology extension plan	SDA end-user confirmation of requirements and formal substrate scope
STOIC extensions	+4	STOIC extensions	STOIC extensions Report approved by SDA Monitor
Task 1 Completion: Formal Substrate Established	+6	Homomorphism Library baseline, Bayesian morphism definitions, provenance ontology	Formal Substrate Design Validation Report approved by SDA Monitor
Task 2 Kickoff: DEVS-PWSA Tracking Layer + Ingestion Pipeline	+8	Tracking Layer SES structure, ingestion pipeline architecture	Experimentation results confirming SES structure and ingestion feasibility
Task 2 Midpoint: Multi-Tool Ingestion Backbone Operational	+10	Working ingestion adapters, model↔graph database bridge, Tracking Layer DEVS models	SDA acceptance of software delivery for Tracking Layer DEVS models

Task 2 Completion: Integrated Tracking Layer Modeling	+12	Fully integrated Tracking Layer DEVS-PWSA environment	integrated Tracking Layer DEVS-PWSA reports and validated use-case results
Task 3 Kickoff: Agentic AI Orchestrator + KPP Reverse-Flow	+14	Initial orchestrator integration with MBSE repositories and graph store	SDA review of orchestrator architecture and data-flow correctness
Task 3 Completion: Automated KPP Reverse-Flow + V&V Traceability	+16	Automated KPP↔design mapping, V&V traceability engine, gap-analysis reports	SDA acceptance of automated reasoning outputs and traceability coverage
Task 4 Kickoff: Scenario Engine + Ontology Maintenance Integration	+18	Unified scenario-execution environment architecture	SDA confirmation of integration plan and environment readiness
Task 4 Completion: Digital Engineering Factory Deployment	+20	Fully integrated scenario engine, ontology pipeline, visualization tools	Successful execution of integrated scenario tests on UA Digital Engineering Factory
Task 5 Kickoff: Bayesian Tranche-Evolution Engine	+22	Bayesian update engine prototype, tranche-evolution sensitivity analysis	SDA review of Bayesian update correctness and morphism-based reasoning
Flagship Demonstration: Tranche-1 → Tranche-2	+23	End-to-end demonstration package, scenario execution, tranche-evolution results	SDA acceptance of flagship demo and transition-ready capability
Final Delivery: Transition Package + Documentation	+24	Final Technical Report, software, training materials, commercialization update	All required reports and documentation confirmed by SDA Monitor

Executive Summary

Space Development Agency (SDA) requires a mission-engineering environment that can evolve at the same pace as the Proliferated Warfighter Space Architecture (PWSA). Each new tranche introduces updated payloads, waveforms, CONOPS, and vendor models, yet SDA must still deliver rapid, defensible assessments of architecture-level performance, resilience, and mission-thread impacts. Current tools—fragmented across performers and inconsistent in fidelity—cannot support the rigor-at-speed posture demanded by tranche-based acquisition.

RTSync's Phase II effort delivers an integrated, mathematically grounded mission-engineering engine purpose-built for continuous PWSA evolution. Building directly on Phase I foundations—including the DEVS-PWSA modeling family, the morphism-based model-comparison framework, the digital-thread methodology, and early agentic-AI orchestration—the Phase II system unifies formal methods, stochastic simulation, automated traceability, and multi-performer model ingestion into a single operational capability hosted on the University of Arizona Digital Engineering Factory (DEF).

The Phase II work is organized into five tightly coupled objectives:

1. **Establish the Formal Substrate** Extend the Systems Theoretic Ontological Integration Core (STOIC) ontology, homomorphism library, Bayesian morphism framework, and provenance ontology into a complete reasoning substrate that supports automated traceability and evidence-weighted updates across evolving tranches.

2. **Expand DEVS-PWSA and Build the Ingestion Backbone** Incorporate Tracking-Layer behaviors and implement a multi-tool ingestion pipeline that unifies MBSE artifacts, simulation models, and graph-based representations, enabling rapid integration of heterogeneous performer models.
3. **Automate KPP Reverse-Flow and V&V Traceability** Deploy an agentic-AI orchestrator that links mission-level KPPs to subsystem-level design choices and test evidence, automatically surfacing traceability gaps and untested requirements.
4. **Integrate Scenario Execution and Deploy the Digital Engineering Environment** Combine the formal substrate, modeling environment, and reasoning components into a unified scenario-execution environment with visualization and ontology-maintenance pipelines.
5. **Deliver the Bayesian Tranche-Evolution Engine and Flagship Demonstration** Implement a Bayesian update engine that maintains confidence in mission performance as new evidence and vendor updates arrive, culminating in a Tranche-1 → Tranche-2 demonstration.

A key innovation is the Bayesian tranche-evolution capability, which addresses a persistent challenge identified in the solicitation: when new tranche data updates only part of the architecture, the system must avoid propagating unintended belief shifts across unrelated subsystems. By combining Bayesian network logic with system-theoretic morphisms, the proposed engine updates confidence only where evidence warrants it—enabling SDA to absorb continuous vendor updates without destabilizing mission-level assessments.

The result is a scalable, auditable, and operationally demonstrable mission-engineering capability that allows SDA to evaluate architecture-level performance and resilience with unprecedented speed and rigor. By unifying formal methods, stochastic simulation, digital-thread automation, and agentic-AI orchestration, this Phase II effort provides SDA with a transition-ready foundation for continuous mission-engineering assessment across future PWSA tranches.

(3) Identification and Significance of the Problem or Opportunity

The need for an integrated mission-engineering capability becomes clear when examining the operational and analytical challenges SDA faces today. SDA is confronting a rapidly intensifying mission-engineering challenge as the Proliferated Warfighter Space Architecture (PWSA) evolves through fast, overlapping tranche cycles. Each tranche introduces new payloads, waveforms, CONOPS, and vendor-specific models, yet SDA must still deliver timely, defensible assessments of architecture-level performance and resilience. Current analysis and VV&A workflows cannot keep pace. As the proposal notes, existing tools are "**fragmented across performers, inconsistent in fidelity, and slow to integrate**," creating delays and uncertainty in mission-thread evaluation.

This challenge is amplified by the operational environment in which PWSA must function. SDA must assess not only nominal performance but also resilience under degraded communications, cyber disruption, node loss, adversarial interference, and cascading multi-domain effects—conditions that often emerge only in rare or low-probability scenarios. This proposal highlights this explicitly: SDA must evaluate resilience **under degraded communications, cyber disruption, node loss, adversarial interference, and cascading multi-domain effects**. These stressors demand modeling and simulation methods that can represent uncertainty, propagate evidence, and quantify confidence in mission outcomes.

Compounding the problem is the continual influx of tranche updates and performer-supplied models. SDA must maintain traceability across mission threads, MBSE artifacts, simulation models, and analysis products while integrating heterogeneous contributions from multiple vendors. Today's workflows—largely document-centric, manually synchronized, and brittle—cannot support the rigor-at-speed posture required for tranche-based acquisition. The result is a widening gap between evolving digital engineering artifacts and the decision-quality mission analysis needed for acquisition, design, and operational planning.

The opportunity, therefore, is to establish a mathematically grounded, computationally scalable, and operationally aligned mission-engineering environment that unifies models, data, and evidence across the SDA ecosystem. A DEVS-based approach is particularly well suited to this need. RTSync's stochastic DEVS methods support rapid model adaptation, heterogeneous model integration, and rare-event simulation—capabilities essential for evaluating large constellations and multi-domain mission threads under uncertainty. With more than two decades of DEVS research, a well-tested simulation engine, and established collaboration with the University of Arizona's Systems Engineering Department, RTSync is positioned to deliver not just models, but a coherent, extensible mission-engineering capability. By addressing the disconnect between evolving engineering artifacts and mission-level decision needs, the proposed Phase II effort enables SDA to perform faster, more rigorous, and more scalable evaluation of future PWSA tranches—ultimately improving acquisition agility, design confidence, and operational readiness.

These challenges motivate the Phase II objectives, which translate the identified needs into a coherent set of technical, ontological, and operational capabilities.

(4) Phase II Technical Objectives

The Phase II effort is designed to transform the foundational methods developed in Phase I into an operational, integrated, and mission-engineering capability for the PWSA prototype ready for evaluation. The work is organized around five mutually reinforcing objectives, each supported by a rigorous technical approach and each offering clear commercial pathways beyond the immediate SDA use case.

Objective 1 — Establish the Formal Substrate for Tranche-Evolution Reasoning

Technical Objective: Extend the Phase I formal methods—including the STOIC ontology family, Homomorphism Library, Bayesian morphism framework, and evidence-type provenance ontology—into a complete substrate that supports automated reasoning, traceability, and evidence-weighted updates across evolving PWSA tranches.

Technical Approach: The team will formalize PWSA-specific mission concepts within the STOIC ontology, instantiate the Homomorphism Library for representative subsystems, and define Bayesian morphisms that quantify structural and behavioral similarity under partial information. A provenance ontology aligned with PROV-O will ensure that all evidence used in tranche-evolution updates is traceable, auditable, and machine-interpretable.

Commercial Applications: This formal substrate has direct applicability to commercial aerospace digital-engineering ecosystems, satellite-constellation integrators, and defense-adjacent industries that require model interoperability, automated traceability, and evidence-based decision support (e.g., autonomous systems certification, safety-critical digital twins).

With the formal substrate established, development moves to the following Objective.

Objective 2 — Extend the DEVS-PWSA Modeling Environment and Build the Ingestion Backbone

Technical Objective: Expand the DEVS-PWSA modeling environment to include Tracking-Layer behaviors and create a robust ingestion pipeline capable of integrating heterogeneous models from multiple performers and tools.

Technical Approach: The team will extend the DEVS-PWSA framework to represent sensor pointing, custody handoff, and missile-warning event chains. A multi-tool ingestion pipeline [5,6,10,16] will be implemented to unify MBSE artifacts, simulation models, and graph-based representations. The automated model↔graph-database bridge will ensure round-trip consistency and enable continuous updates as new models arrive.

Commercial Applications: The ingestion backbone is directly applicable to commercial digital-engineering platforms, model-based systems engineering toolchains, and industries that must integrate models from multiple vendors (e.g., automotive Automated Driver Assistance System ecosystems, aerospace supply chains, energy-grid digital twins).

With ingestion and modeling in place, development moves to the following Objective.

Objective 3 — Deliver Automated KPP Reverse-Flow and V&V Traceability

Technical Objective: Operationalize automated reasoning that links mission-level Key Performance Parameters (KPPs) to subsystem-level design choices and test evidence, enabling continuous assessment of requirements coverage and system readiness.

Technical Approach: The team will implement an agentic-AI orchestrator that coordinates simulation tools, graph stores, and MBSE repositories. Bidirectional KPP↔design-parameter mapping will be automated using the Mission Thread Ontology and the digital-thread methodology developed in Phase I. The system will surface traceability gaps, untested requirements, and morphism-based degradation.

Commercial Applications: Automated traceability and requirement-to-design reasoning are in high demand across commercial aerospace, medical-device certification, automotive safety compliance, and any regulated industry that must demonstrate end-to-end verification and validation.

These reasoning capabilities feed directly into the integrated scenario-execution environment.

Objective 4 — Integrate Scenario Execution and Deploy the Digital-Engineering Environment

Technical Objective: Integrate the formal substrate, modeling environment, and automated reasoning components into a unified scenario-execution and visualization environment hosted on the University of Arizona DEF.

Technical Approach: The team will integrate scenario execution, visualization, and ontology-maintenance pipelines into a cohesive environment. Continuous validation gates will ensure that ontology updates, model changes, and evidence ingestion remain consistent. The system will be deployed on the UA DEF to support SDA evaluation and multi-domain harmonization.

Commercial Applications: Scenario-execution environments are broadly applicable to commercial mission-analysis tools, enterprise digital-twin platforms, and training/simulation environments used in aerospace, defense, energy, and logistics.

With the environment unified, Phase II delivers the Bayesian tranche-evolution engine.

Objective 5 — Deliver the Bayesian Tranche-Evolution Engine and Flagship Demonstration

Technical Objective: Produce a fully operational tranche-evolution engine that updates mission-engineering confidence as new evidence, vendor models, and tranche-scope changes arrive, culminating in an end-to-end Tranche-1→Tranche-2 demonstration.

Technical Approach: The team will operationalize the Bayesian morphism framework, perform tranche-evolution sensitivity analysis, and execute the flagship demonstration using the integrated environment from Tasks 1–4. The final deliverables include the tranche-evolution engine, demonstration package, software and documentation. Attention will be paid to reducing the learning curve for, and increasing the user-friendliness of, the transferred software and its documentation in recognition of the complexity for non-PhD SDA users.

Commercial Applications: Evidence-weighted evolution engines have strong commercial potential in satellite-constellation management, autonomous-system lifecycle assurance, predictive maintenance, and any domain where system configurations evolve over time and require continuous confidence assessment.

Summary

Together, these objectives deliver a rigorous, integrated, and operationally relevant mission-engineering capability for SDA. The technical approach combines formal methods, stochastic simulation, digital-thread automation, and agentic-AI orchestration into a unified environment that supports rapid, evidence-based tranche evolution. Each objective also offers clear commercial pathways, ensuring that the Phase II effort provides both immediate value to SDA and long-term applicability across the broader digital-engineering ecosystem.

(5) Work Plan

1. Objective

The objective of the Phase II effort is to extend, operationalize, and integrate the formal, ontological, simulation, and agentic-AI foundations established in Phase I into a coherent mission-engineering and tranche-evolution capability for the Proliferated Warfighter Space Architecture (PWSA). The contractor shall deliver a mathematically grounded, digitally integrated, and operationally relevant engine that supports SDA's rigor-at-speed posture and enables continuous evaluation of evolving PWSA tranches. In more detail the objectives are 1) **Establish the formal substrate** (ontology extensions, morphisms, Bayesian reasoning, provenance). 2) **Extend DEVS-PWSA modeling and build the ingestion backbone** for multi-performer integration. 3) **Automate Key Performance Parameters (KPPs) reverse-flow and V&V traceability** using AI-driven orchestration and digital-thread methods. 4) **Integrate scenario execution and deploy in a capable digital-engineering environment**. 5) **Deliver the Bayesian tranche-evolution engine and flagship demonstration** for Tranche-1→Tranche-2 transition readiness.

2. Scope

The Phase II scope is structured to build directly on the validated artifacts, prototypes, and methods delivered in Phase I. The contractor shall extend the Phase I DEVS-PWSA modeling family, the morphism-based model-comparison framework, the digital-thread methodology, and the agentic-AI orchestration concepts into a fully integrated capability deployable, as a prototype, on the University of Arizona DEF. The proposed workflow to be presented below summarizes the Phase II execution sequence aligned to the above objectives. Task 1 establishes the formal substrate. Task 2 extends the modeling environment and builds the ingestion backbone. Task 3 implements automated KPP reasoning and V&V traceability. Task 4 integrates these

components into a unified scenario-execution and digital-engineering environment. Task 5 delivers the tranche-evolution engine and the flagship demonstration.

3. Background

a. Introduction

Current practices followed by SDA tend to rely on document-based practices that are static and siloed, and impede the desired speed, delivery, and agility. As indicated above, to deliver capabilities and systems to the desired speed and with the desired agility, SDA must look to data-centric approaches that are transformative in nature. The defense industry began its digital transformation in earnest after the release of the digital engineering strategy in 2018 [24]. Although digital methodologies have been around for decades, true transformation is still emerging. Even within the same set of tools, different organizations often have varied implementations, leading to a lack of reuse and interoperability. This challenge is particularly pronounced in mission engineering, where interdependent organizations must collaborate to develop, operate, and maintain systems and capabilities suitable for warfighters. For instance, models created in model-based systems engineering languages and tools, such as Systems Modeling Language (SysML) and Cameo, often adhere to a "style guide" that is organization-specific [4,29]. Overcoming these obstacles is a crucial step in achieving reuse and interoperability, which are essential for speed and agility.

As the complexity of systems and capabilities grows, so does the complexity of the models that represent them. Are these models intended for human analysis or for computer analysis that is then interpreted for humans? Human consumption of complex models is challenging. Even in less complex scenarios, SysML and Cameo models can be interpreted in various ways [2,9,29,39]. Additionally, Model Based System Engineering (MBSE)[14,32,39] lacks a common underlying mathematical theory, unlike computational fluid dynamics and finite element analysis. While SysMLv2 has ontological underpinnings, it lacks a mathematical foundation. To manage the complexity of missions, capabilities, systems, and models, SDA should seek a digital framework supported by a robust mathematical theory.

True digital transformation involves making effective and efficient decisions and communication by connecting the right data correctly. Two mathematical frameworks have been proposed for integration to support this transformation [32]: *the Tricotyledon Theory of System Design* (T3SD) [36], known for coining the term MBSE [36], and the Theory of Modeling & Simulation (commonly referred to as *Discrete Event System Specification*, or *DEVS*) [42]. The advancement of MBSE, grounded in T3SD, is gaining momentum, while DEVS is widely recognized in the modeling and simulation community, with over 10,000 publications. Our project aims to apply research towards integrating T3SD and DEVS into the space domain, specifically to achieve digital test and evaluation (dT&E) [8] for the SDA.

b. DEVS Background

The DEVS framework stands out for its hierarchical and modular architecture. Within this structure, a system or System-of-Systems (SoS) model consists of several individual models, each representing a system component. These models are classified as either atomic (indivisible) or coupled (comprising other models). DEVS distinctly treats agent-based and other simulation elements as specific atomic or coupled models, complete with their state transitions, inputs, and outputs. By adopting a systems-theory approach, DEVS ensures the integrity of each simulator's fidelity, be it high-resolution or a more generalized representation, while fostering their collaboration. The versatility of the DEVS framework is further emphasized by its support for varied M&S paradigms [42]. Despite the potential differences in atomic or coupled models—like differing state variables or timing mechanisms—the overarching DEVS simulation protocol ensures their correct interaction. This adaptability not only allows DEVS to model and simulate a diverse range of systems but also facilitates the amalgamation of independently crafted models, promoting efficiency and reusability.

Agent-based modeling is among the paradigms specifically supported by RTSync's DEVS Markov modeling capabilities [3,25,26,35,44]. DEVS Markov models can be individualized with specific transition probabilities and transition times that are reconfigurable during model execution for dynamic structural change – ideal for

agent-based decision making. *DEVS Markov models are full-fledged DEVS models and can be coupled with other DEVS components in hierarchical compositions [11,20,41,42] to unravel the complexities of interconnected agents needed to analyze and evaluate space-enabled kill-webs [1,7,15,21].*

c. System Entity Structure (SES) Background

The **System Entity Structure (SES)** is a formal framework to support **model organization, reuse, and configuration** in the theory of modeling and simulation [25,26,37,41,42]. It provides a **tree-structured ontology** that captures the structural knowledge of a system in a modular and hierarchical way. SES was designed to manage the complexity of modeling large or distributed systems by separating **structural knowledge** (what entities exist and how they can be combined) from **behavioral models** (how those entities act over time). The SES defines a **family of possible system configurations** rather than a single model. Through **pruning operations**, specific models can be derived from the general structure, enabling systematic exploration of alternatives. Key Features are a) **Hierarchy**: Supports recursive decomposition of systems into subsystems. b) **Abstraction**: Allows reasoning at multiple levels of detail. c) **Reusability**: Families of models can be generated without redefining structures. d) **Integration with DEVS**, where SES provides the structural blueprint and DEVS defines the dynamics.

d. Experimental Frames (EF) Background

The SES ties into **Experimental Frames (EFs)** and **Verification & Validation (V&V)** workflows: An Experimental Frame specifies *how* a system will be observed, stimulated, and measured (inputs, outputs, metrics, conditions) [42,46]. SES defines the **space of candidate models** While EF defines the **conditions under which those models are tested**. Together, they allow systematic derivation of *model–experiment pairs* for simulation studies. In model-based T&E, the SES can generate alternative system configurations, while EFs provide the test harnesses (stimuli, monitors, performance metrics) to yield executable test scenarios. By pairing SES-derived models with EFs, one can test whether the model's behavior matches stakeholder intent [27,28,30]. Multiple variants from SES can be validated against the same EF to see which structural assumptions best align with real-world data. This supports *comparative validation*—evaluating families of models rather than a single candidate. SES + EF provides a **formal bridge** between *model generation* and *model evaluation*. It supports **systematic exploration** of design alternatives while embedding **V&V discipline** into the modeling workflow. This approach scales well for **complex systems-of-systems**, where both structure and context vary.

Together, DEVS, SES, and EF provide the mathematical and structural foundations that Phase II builds upon. The following Technical Approach Overview shows how these form the substrate for PWSA mission-engineering, ingestion, scenario execution, and tranche-evolution reasoning.

e. Technical Approach Overview

The PWSA demands a mission-engineering environment capable of evolving at the same pace as its tranche-based acquisition model. DEVS-PWSA provides this capability through a mathematically rigorous, model-agnostic simulation engine built on DEVS and its stochastic extension [13]. This foundation enables SDA to evaluate new capabilities, payloads, and operational concepts as soon as they emerge, rather than waiting for lengthy re-integration cycles. The architecture is expressly designed to support rapid iteration, heterogeneous model integration, and automated verification—capabilities essential for a dynamic, distributed, and continuously evolving space enterprise.

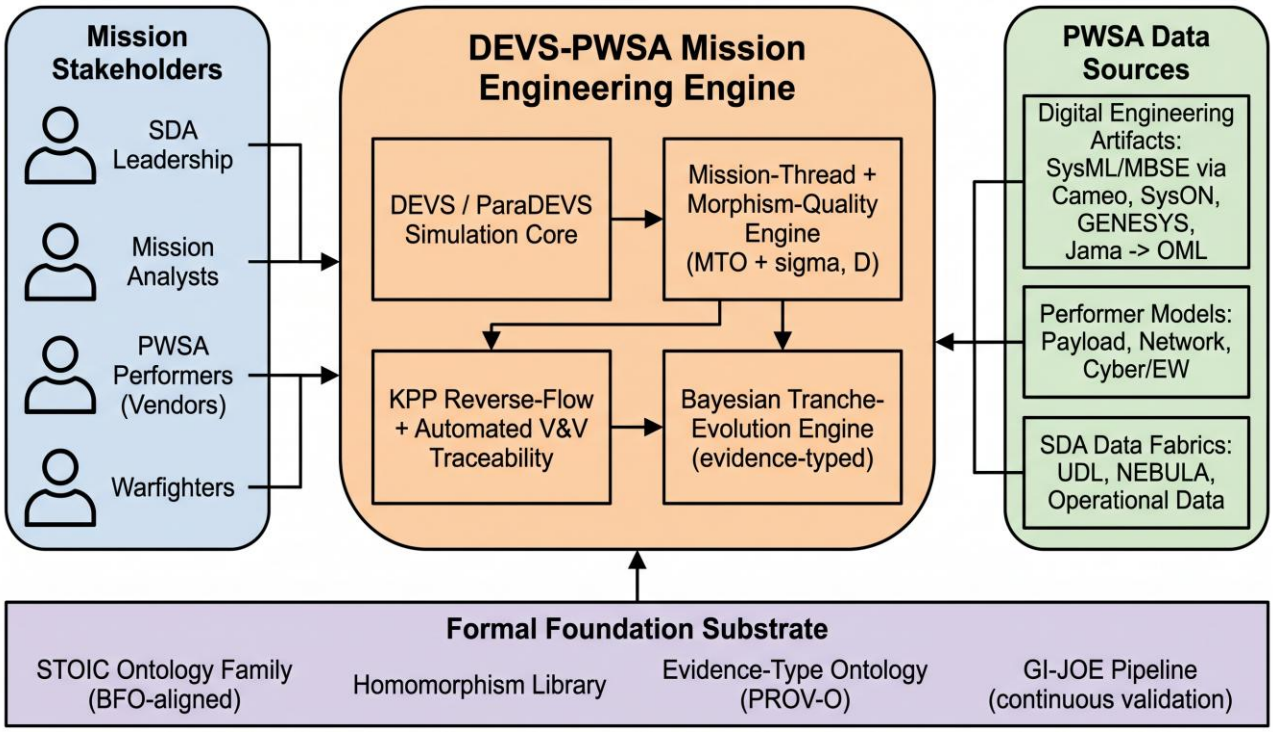


Figure 1. *DEVS-PWSA Architecture and Mission Engineering Engine*

Figure 1 shows the core of the approach, RTSync's DEVS-based state-transition engine, which provides a formal, composable representation of system behavior. Unlike ad-hoc or discrete-time simulators, DEVS enforces strict semantics for state, events, and timing, ensuring that every model—whether representing a satellite, ground node, link, or mission thread—executes with mathematical consistency. This formalism enables **rapid tranche-to-tranche evolution** by allowing new models to be inserted, replaced, or updated without destabilizing the broader architecture. Automated V&V routines detect inconsistencies as models evolve, ensuring that SDA maintains a continuously validated mission-engineering environment.

A second pillar of the approach is high-fidelity rare-event and resilience analysis. PWSA's operational value depends on its ability to withstand link degradation, cyber disruptions, node failures, adversarial interference, and multi-domain cascading effects. Traditional Monte Carlo or discrete-time tools often miss these low-probability, high-impact behaviors. DEVS-PWSA incorporates a stochastic simulation engine capable of capturing rare-event dynamics with precision, revealing vulnerabilities that only appear under extreme or adversarial conditions. This gives SDA early insight into resilience gaps and informs the design of more robust architectures, kill chains, and mission threads.

The third component of the technical approach is seamless integration with SDA's digital-engineering ecosystem. DEVS-PWSA is engineered to interface with the **Unified Data Library** (UDL), NEBULA, and emerging SDA data fabrics, enabling direct ingestion of authoritative data sources. The framework also consumes SysML, MBSE artifacts, and mission-thread definitions, ensuring that digital-engineering products flow directly into mission-level analysis without translation loss. External simulation tools, operational data streams, and performer-developed models can be integrated through standardized interfaces, positioning DEVS-PWSA as the unifying mission-engineering layer across the performer ecosystem.

Scalability is a fourth defining feature. PWSA's proliferated architecture requires simulation of hundreds to thousands of nodes across multiple domains. DEVS-PWSA is optimized for high-performance execution,

supporting real-time and faster-than-real-time simulation of large-constellation scenarios. Parallel execution, efficient event scheduling, and distributed simulation capabilities enable rapid exploration of architectural variants, resilience strategies, and mission-thread alternatives. This computational efficiency allows SDA to evaluate hundreds of design options within a single tranche cycle, accelerating decision timelines and reducing engineering overhead.

The technical approach also leverages RTSync's mature technology base and proven transition record. With more than two decades of DEVS research, multiple Phase II successes, and operational deployments across defense programs, RTSync brings a production-ready simulation engine rather than a prototype. The company's long-standing partnership with The University of Arizona's Department of Systems and Industrial Engineering further strengthens the research foundation and ensures alignment with national-security mission needs. This maturity reduces technical risk and ensures that Phase II outputs are deployable capabilities ready for SDA integration.

DEVS-PWSA is designed to complement, not replace, the broader performer ecosystem. While other performers may focus on hardware integration, transport-layer communications, operator training, cybersecurity tooling, or data-fusion pipelines, DEVS-PWSA provides the mission-engineering substrate that unifies these contributions [17,18]. The framework ingests outputs from diverse technical domains and converts them into **quantitative performance assessments**, end-to-end mission-thread evaluations, resilience metrics, and architecture-level trade studies. This ensures SDA evaluates the system-of-systems, not just individual components.

Envisioned Operational Use Cases Motivating Phase II Objectives

To illustrate how DEVS-PWSA will be used in practice, the following operational scenarios demonstrate the kinds of rigor-at-speed mission-engineering activities SDA must perform as the PWSA evolves. These use cases reflect real decision pressures faced by SDA analysts and acquisition teams and show how the Phase II capability directly supports tranche-based development, vendor integration, and mission-thread assessment.

- **Rapid Vendor Model Update Integration During Tranche Cycles** As vendors deliver late-cycle updates to payload, waveform, or autonomy models, SDA analysts ingest the new artifacts through the multi-tool pipeline. The morphism engine quantifies structural and behavioral differences from prior versions, and the Bayesian tranche-evolution engine updates confidence only for the subsystems affected by the change. Within hours—not weeks—SDA can see which KPPs shift, which remain stable, and whether the update introduces new risks. This workflow motivates the ingestion backbone (Objective 2) and the Bayesian update engine (Objective 5).
- **Custody Continuity Assessment in the Tracking Layer** When SDA evaluates a new sensor-pointing strategy or custody-handoff algorithm, analysts select the relevant mission thread (e.g., missile warning) and execute Tracking Layer DEVS models under nominal and degraded conditions. Experimental Frame monitors detect custody breaks, latency spikes, and loss-of-track events. The Bayesian engine updates confidence in custody-related KPPs, enabling SDA to quantify whether the new algorithm improves resilience. This motivates the Tracking Layer modeling extensions (Objective 2) and scenario-execution environment (Objective 4).
- **On-Orbit Acceptance Support for Vendor Digital Twins** Prior to hardware launch, SDA must determine whether a vendor's digital twin is sufficiently representative of expected on-orbit behavior. The morphism-quality framework computes structural and behavioral distance between the vendor's model and the authoritative reference. The system classifies the twin as Ideal, Coarse-but-Safe, Precise-but-Wrong, or Divergent, and the Bayesian engine weights the vendor's evidence accordingly. This motivates the formal substrate (Objective 1) and the tranche-evolution engine (Objective 5).

- **Mission-Thread Impact Analysis Under Node Loss or Cyber Disruption** SDA routinely evaluates resilience under degraded conditions such as node loss, RF interference, or cyber-induced link degradation. Analysts configure a degraded-mode scenario, execute multi-domain DEVS models, and observe mission-thread impacts on latency, coverage, and continuity. The Bayesian engine updates mission-level confidence and highlights which subsystems drive the degradation. This motivates the scenario-execution environment (Objective 4) and the Bayesian update engine (Objective 5).
- **Automated KPP Reverse-Flow for Requirements Coverage and Gap Analysis** When SDA needs to understand which subsystem design choices influence a specific KPP—such as detection latency or custody duration—the reverse-flow engine traces dependencies through the Mission Thread Ontology and identifies untested requirements, missing evidence, and design parameters with outsized impact. This motivates the automated KPP reasoning and V&V traceability capability (Objective 3).
- **Cross-Vendor Comparison and Architecture-Level Trades** SDA often receives competing models from multiple vendors for the same subsystem. The morphism engine provides a quantitative comparison of structural and behavioral fidelity, while the scenario engine evaluates mission-thread performance under each vendor's assumptions. This motivates the ingestion backbone (Objective 2) and the formal substrate (Objective 1).

Together, these envisioned use cases demonstrate how DEVS-PWSA enables SDA to perform rapid, defensible, and evidence-driven mission-engineering assessments across evolving tranches. They also directly motivate the Phase II objectives: establishing the formal substrate, extending the modeling environment, automating KPP reasoning, integrating scenario execution, and delivering the Bayesian tranche-evolution engine.

These operational scenarios directly motivate the Phase II objectives, each of which addresses a specific capability required to support SDA's tranche-evolution mission-engineering workflows.

In the proposed work, the formal mathematical foundation of DEVS provides the basis for maintaining traceability across evolving requirements. Because system behavior, interfaces, and timing are represented within a well-defined formalism, new payloads, waveforms, or CONOPS can be incorporated through modular model updates while preserving architectural coherence. This same formal basis supports automated V&V routines that identify interface mismatches, timing inconsistencies, and behavioral conflicts as the model family evolves. That traceability is essential for tranche-based acquisition, where rapid iteration must be balanced with rigorous validation.

The proposed approach also uses this formal representation to support transparency and explainability. By representing internal state transitions, timing logic, and event dependencies explicitly, the framework enables analysts to understand why a mission-thread outcome occurred, not just what the outcome was. In the proposed implementation, this supports mission assurance, operational planning, and senior-leader decision support by enabling scenario replay, internal-state inspection, and comparison of alternative architectures.

Exploratory Analysis and Operational Decision Support

Another important aspect of the proposed approach is that the formal mathematical foundation of DEVS supports both exploratory analysis and operational decision support within a single modeling environment. In Phase I, that formal representation was used to explore architectural options, resilience strategies, and kill-chain alternatives in a consistent way.

Model	Model Feature	Parameters and Relations
PWSA Operations Model	• Describes operation at the System of Systems level with component systems such as Satellite Operations, Mission Processing, Battle Management, Ground Support, and Communication Network. • Component systems provide specific services and engage in message exchanges such as mission directives and tasking, mission status updates and resource allocation.	
Base Model	Physics-based characterization of constellation structure and environment	Conceptual comprehensive system representation
Stochastic Connectivity	Predicts optimal message transmission paths under physical conditions mapped from Base Model with variance in design parameters	Random variables for connection parameters. E.g. PathDistance = N(mean, sigma)
Lumped Connectivity Model	Predicts expected message transmission paths under physical conditions mapped from Base Model	Utilizes lumped (average) values for connection variables E.g. PathDistance = <PathDistance>
Markov Model	• Stochastic model predicts SoS level statistics concerning message transmission quality (connectivity and latency)	Lumps connection variables into Probability of connection E.g. PathDistance not utilized

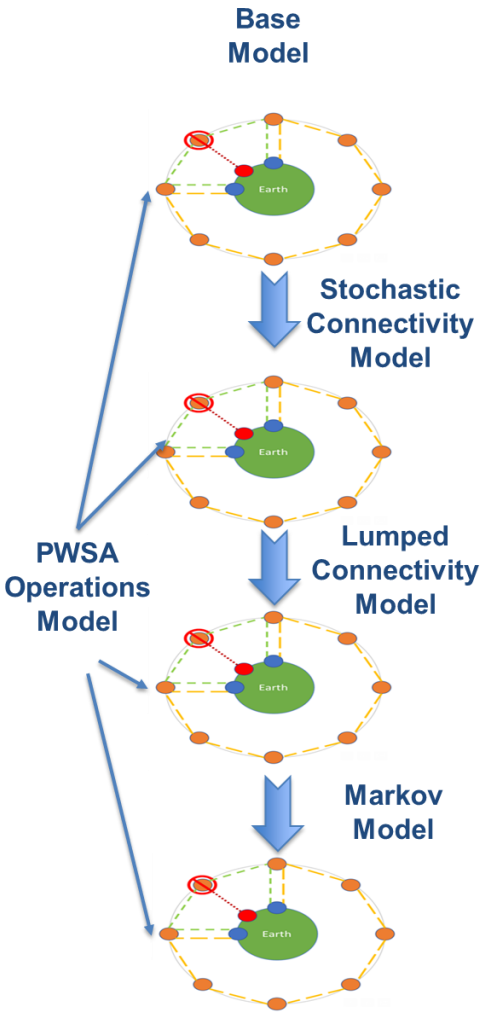


Figure 2. *Model Chain for Solar-Flare Feasibility Demonstration*

Figure 2 shows how the solar-flare scenario is represented across the family of models linked by morphisms. The **Base Model** captures the physical environment and satellite geometry; the **Stochastic and Lumped Connectivity Models** translate those conditions into varying or averaged communication paths; the **Markov Model** summarizes overall link reliability and latency; and the **PWSA Operations Model** integrates these results at the mission level. Together, the chain demonstrates how transient solar-flare effects propagate from physical disturbance to system-of-systems performance.

The Phase I feasibility demo explored how a **solar flare** could temporarily disrupt satellite communications along a normal message path. The physics-based Base Model in Figure 2 characterized how such a flare increases atmospheric ionization and noise in radio frequencies, which can reduce signal quality and slow message transmission. In a simplification process that was formalized with mathematical morphism [38,42,45] concepts implemented in database ontological relations, reduced order lumped models captured the disruptive effects on communication and enabled engineering mitigations to be studied. Together, this model family

demonstrated how environmental events like solar flares can be integrated into the broader **PWSA Operations Model** for mission-level analysis.

In Phase II and subsequently, the system matures, the same representation can support operational planning, contingency analysis, and real-time what-if assessments without requiring a separate analytic framework. In the proposed implementation, this continuity supports use of the environment across the system lifecycle.

Finally, the approach is designed for long-term sustainability because the proposed implementation uses DEVS and its support for the features introduced in Section 3) (modular architecture, standardized interfaces, and automated V&V pipelines [4, 22, 23, 29]). In this context, formal representation helps reduce the cost of maintaining and evolving the mission-engineering environment as models, interfaces, and requirements change over time. As SDA's architecture grows in scale and complexity, this provides a stable and extensible basis for continued mission-level evaluation. Realization in the MS4 Me simulation environment utilized DEVS models coupled via ports to follow specific message flows to visualize the performance degradation under stressed conditions.

The ontology-engineering substrate provides the machine-interpretable structure over which Bayesian confidence propagation operates as detailed now.

i. Bayesian and Systems-Theoretic Methods for V&V Confidence

The Phase II framework propagates Verification and Validation confidence across PWSA tranche evolution using Bayesian networks structured by the Mathematical Systems Engineering Metamodel (Figure 3). The motivation is operational: the Space Development Agency's (SDA) 2026 rigor-at-speed and vendor on-orbit acceptance posture creates a need for quantified confidence on Key Performance Parameter (KPP) attainment under continuously updated information, rather than periodic point-estimate verifications. Traditional V&V approaches treat each evidence event as a discrete pass-or-fail outcome; the proposed framework treats them as observations that update posterior beliefs about model fidelity, subsystem behavior, and mission-thread reliability.

The mathematical foundation rests on the Mathematical Systems Engineering Metamodel and its hierarchy of system specifications introduced earlier. Each node in the Bayesian network represents a stage of PWSA system evolution, including analytical models, lower-fidelity simulations, higher-fidelity simulations, digital twins, and fielded operational systems. Edge probabilities between linked nodes are parameterized by morphism quality, the formal measure of equivalence between the linked specifications. New test data, vendor model updates, and tranche scope changes are absorbed as evidence at the appropriate node, and the Bayesian update propagates the resulting belief shifts through the network according to standard d-separation reasoning [12]. The d-separation treatment is load bearing for tranche evolution: when a tranche update lands data on some subsystems and not others, the belief propagation must respect the conditional independence structure implied by the morphism graph, rather than propagating updates indiscriminately.

This approach builds directly on the team's prior work presented earlier on **conjoining Bayesian methods and systems theory** for the Test and Evaluation of learning-based systems [12,19] and extends the team's draft work on Bayesian and systems-theoretic methods for AI-based systems V&V confidence to the PWSA tranche-evolution context [33,34]. The **proposed Phase II contribution** is to *operationalize* this Bayesian–systems-theoretic confidence framework within the program's actual modeling ecosystem—specifically by grounding it in the team's **Mission Thread Ontology**, the **Homomorphism Library** to be developed under Task 1.2, and the **PWSA-specific scenario evidence** to be generated across Tasks 2 through 4. This represents a shift from conceptual formulation to executable capability: the Bayesian network structure, its

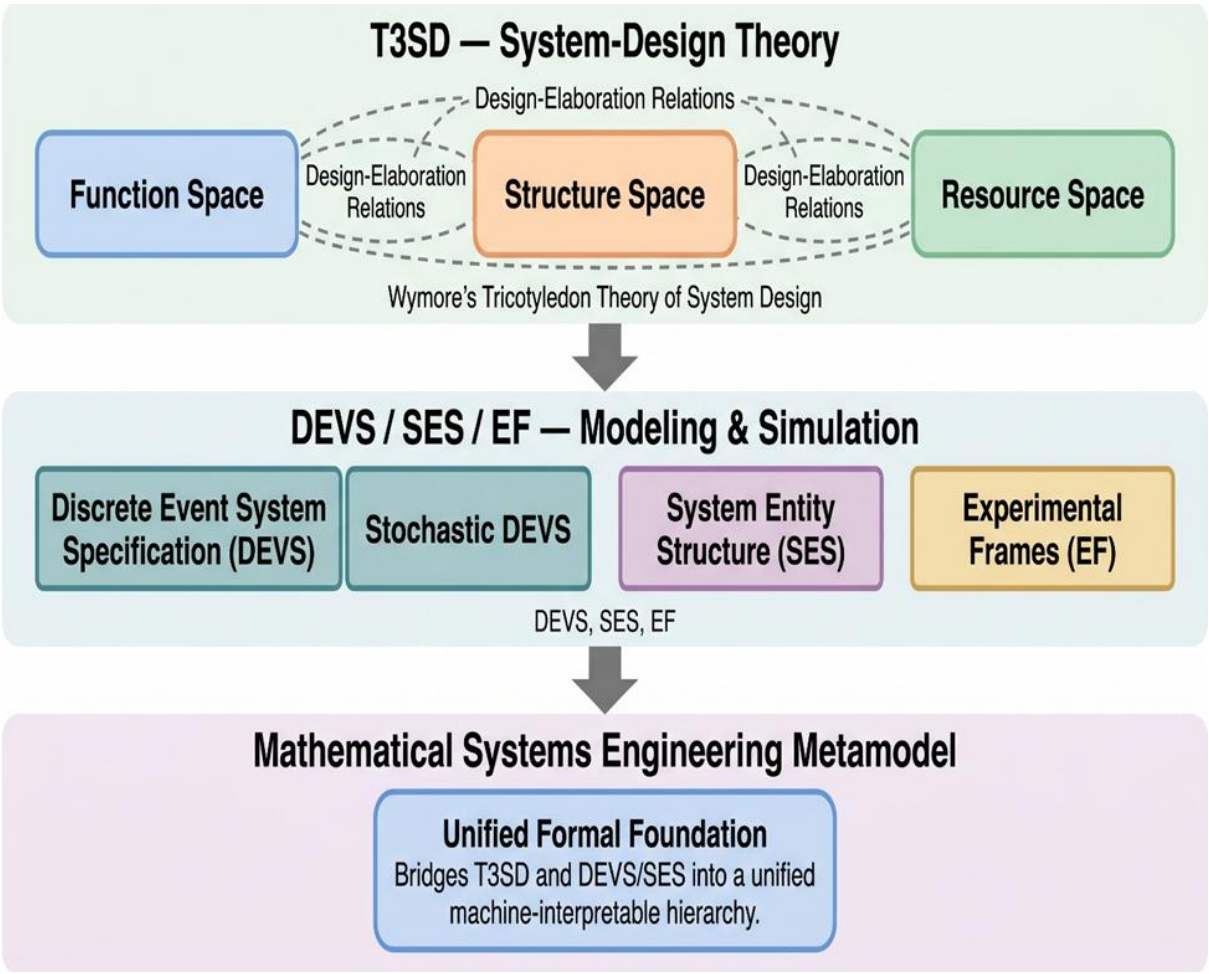


Figure 3 *illustrates how the STOIC ontology family organizes the formal foundations for Phase II by aligning T3SD's Function, Structure, and Resource spaces with their corresponding DEVS/SES constructs. STOIC's Math Systems Theory-based metamodel provides the mathematical bridge that ensures functional requirements, structural architectures, and resource constraints remain consistent and machine-interpretable across all DEVS models, mission threads, and tranche-evolution reasoning workflows.*

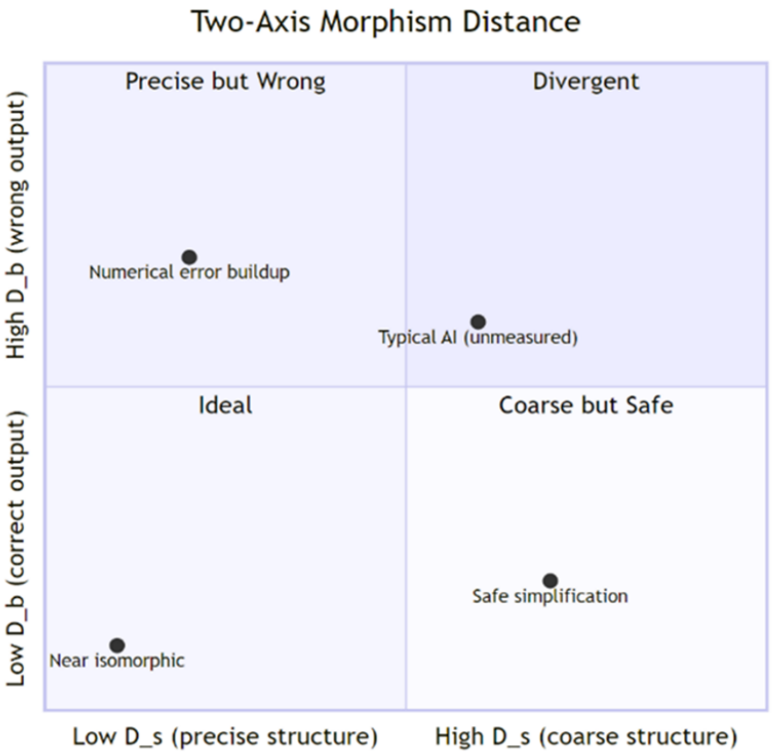
morphism-parameterized edges, and its d-separation–based information-flow rules become instantiated directly in the evolving model family used for tranche-evolution analysis.

To demonstrate feasibility, the team recently conducted an independent assessment exercise using an authentic third-party mission-engineering model. That study produced **empirical morphism-quality measurements** across seven subsystems, showing that the proposed methodology can be applied to *real defense-domain data* rather than only notional examples. This evidence supports the Phase II plan to transition the approach from research to an operational toolchain that provides SDA with mathematically grounded, continuously updated V&V confidence across evolving tranches.

The regulatory landscape further motivates the approach. The National Institute of Standards and Technology Artificial Intelligence (AI) Risk Management Framework establishes Test, Evaluation, Verification, and Validation as a core measurement function. The Department of Defense Digital Test and Evaluation for AI Guidebook emphasizes methods that leverage lower-cost digital testing to inform higher-fidelity operational

testing. The proposed Bayesian and systems-theoretic methods satisfy both directives by **providing a mathematically rigorous mechanism for propagating evidence from lower-fidelity testing to higher-fidelity prediction, with quantified confidence intervals at each stage.**

ii. Morphism Quality Framework: Structural and Behavioral Preservation



Quadrant	Representative PWSA Use Cases
Ideal	Validating a vendor-supplied Tracking Layer sensor model against the government reference model. • Swapping a high-fidelity intersatellite link (ISL) model with a reduced-order version for faster Monte Carlo runs while preserving timing and packet-error behavior. • Comparing Tranche-1 and Tranche-2 payload models where only minor waveform updates occur.
Precise but Wrong	Same satellite bus and pointing architecture, but new waveform or coding scheme changes link performance. • Tracking Layer custody-handoff logic preserved, but vendor-specific detection latencies differ significantly. • Same SES structure for a ground-to-space chain, but new atmospheric conditions (e.g., scintillation) produce different behavior.
Coarse but Safe	Replacing a detailed orbital-propagation model with a lumped-drag model that still reproduces pass-timing behavior for mission-thread analysis. • Using a simplified RF link model that matches SNR-vs-range behavior without modeling antenna-level details. • Aggregating multiple Tracking Layer nodes into a single "effective sensor" for high-level custody timelines.
Divergent	Attempting to reuse a commercial GEO communications model for LEO PWSA timing-critical custody operations. • Using a terrestrial 5G NTN model that does not match PWSA's intersatellite geometry or timing. • Applying a legacy missile-warning model with incompatible detection logic and no structural alignment to Tracking Layer sensors.

Figure 4 *Two-axis morphism distance. StructuralDistance on the x-axis captures how many distinct real-world states or inputs are lumped together by the lower-fidelity specification. BehavioralDistance on the y-captures worst-case output discrepancy. Both axes are zero at an exact isomorphism. The diagonal opposite quadrant, both axes high, represents model construction approaches with neither good structural nor behavioral correspondence to operational reality of interest.*

Morphism quality between linked system specifications is captured on two orthogonal axes (Figure 4). Structural distance, denoted *StructuralDistance* on the x-axis, measures the extent to which distinct real-world

states or inputs are lumped together by the lower-fidelity model. Behavioral distance, denoted *BehavioralDistance* on the y-axis, measures the worst-case output discrepancy between linked specifications under a defined input distribution. Both axes are zero at an exact isomorphism, and the four quadrants of the resulting plane have distinct interpretations.

The *Ideal quadrant* (low *StructuralDistance*, low *BehavioralDistance*) corresponds to near-isomorphic mappings: both structure and behavior are preserved across the linked specifications. The *Coarse but Safe* quadrant (high *StructuralDistance*, low *BehavioralDistance*) corresponds to safe simplification, sometimes called a lumped-parameter model in formal literature: here structure is intentionally simplified, but the output behavior remains close to the higher-fidelity reference [25, 37,42, 45]. The *Precise but Wrong* quadrant (low *StructuralDistance*, high *BehavioralDistance*) corresponds to numerical-error buildup, sometimes called discretization failure in the formal literature: structure is preserved, but accumulated numerical error has displaced the output trajectory. The *Divergent quadrant* (high *StructuralDistance*, high *BehavioralDistance*) is the operationally concerning region; this is where unmeasured artificial intelligence systems typically land, with neither structural correspondence to the operational reality nor bounded behavioral agreement.

To ensure that morphism-quality metrics remain computationally tractable even when comparing stochastic DEVS models, Phase II incorporates the ParaDEVS [47] branching-time tree-expansion method, which directly addresses the combinatorial explosion inherent in evaluating behavioral distances over large stochastic spaces. By integrating this merging-based tree-expansion framework into the morphism-quality engine, Phase II ensures that structural and behavioral distance metrics remain scalable, accurate, and efficient even for deeply branching stochastic DEVS models. Behavioral distance is computed as a conservative upper bound rather than an exhaustive supremum: the ParaDEVS merging step prunes behaviorally equivalent branches, and the worst-case output discrepancy is estimated over inputs sampled from the experimental-frame distribution, with the Task 1 partial-observation treatment governing states that are not directly observed.

The framework supports two operational uses in Phase II. First, the framework quantifies the fidelity of each layer of the family of models from Phase I and its extension in Phase II, so that the Bayesian tranche-evolution engine can weight evidence from each layer appropriately. Second, the framework supports vendor on-orbit acceptance by giving SDA an explicit measure of how far a vendor's digital twin lies from the authoritative reference, before the vendor's hardware reaches on-orbit checkout.

iii. Ontology-Engineering Substrate

As indicated earlier, Phase II builds on a formal ontology-engineering substrate that supports machine-checkable knowledge representation, continuous validation, and traceability across the proposed Tasks. The substrate is grounded in Basic Formal Ontology (BFO) version 2020, the International Organization for Standardization 21838-2 upper ontology, with mid-level alignment to the Common Core Ontologies. This grounding ensures that the PWSA-specific ontology classes to be added during Phase II compose cleanly with the broader defense-domain ontology ecosystem.

As depicted above in Figure 3, the Systems Theoretic Ontological Integration Core (STOIC) ontology family was built upon three complementary layers. The first carries Wymore's Tricotyledon (visualized as 3-leafed sprout) Theory of System Design into a machine-checkable form, supplying the 3 working spaces (Function, Structure, Resource) and design-elaboration relations that underpin Phase I's family-of-models structure. The second carries Discrete Event System Specification formalism, including stochastic DEVS, the System Entity Structure (SES)[40,42,43], and Experimental Frames, supporting the stochastic simulation uncertainty quantification analysis demonstrated in Phase I. The third carries a broader Mathematical Systems Engineering Metamodel, bridging the cotyledon spaces and DEVS into the unified formal foundation that the Bayesian framework reasons over.

For mission-thread representation, Phase II adopts the team's Mission Thread Ontology, an artifact developed and computationally validated in the team's recent Semantic Technologies for Intelligence, Defense, and Security (STIDS) Conference work. In more detail, the Mission Thread Ontology comprises a vocabulary of seventeen classes and eighteen object properties aligned with BFO and the Common Core Ontologies, a ten-shape Shapes Constraint Language (SHACL) graph for closed-world validation, and a twelve-question competency-question catalog. The ontology was independently validated against an authentic third-party mission-engineering model, with logical consistency confirmed under two reasoners and shape conformance confirmed with zero violations. The Phase II contribution extends the Mission Thread Ontology with PWSA-specific concepts and integrates it as the operational mission-thread layer in Tasks 3 and 5.

Ontology maintenance and continuous validation are supported by a multi-agent ontology-engineering pipeline developed in the team's prior research line. The pipeline combines a Web Ontology Language standards stack with the Ontological Modeling Language family, an alignment specialist that maps candidate domain terms to BFO and the Common Core Ontologies, an evaluator that drives a five-layer validation pipeline including SHACL conformance and competency-query coverage, and a two-tier gate that runs advisory syntax checks on every change and a blocking full validation at merge time. The pipeline has been applied to five distinct Systems Engineering knowledge-engineering projects with consistent quality-metric performance. Phase II deploys the pipeline as the continuous-validation gate for all Phase II ontologies in Task 4.2.

The Technical Approach establishes the conceptual and mathematical basis for the Phase II effort. The following Task Requirements translate this foundation into concrete activities, deliverables, and integration milestones aligned with SDA's tranche-based development model.

4. Task/Technical Requirements

Task Plan Outline

Efforts across tasks shall be harmonized for efficient execution and to ensure that each task's outputs serve as inputs to the next stage of the workflow (Figure 5). The contractor shall:

- Establish the formal substrate required for all downstream automation (Task 1).
- Extend the DEVS-PWSA modeling environment to the Tracking Layer and implement the multi-tool ingestion pipeline (Task 2).
- Implement the agentic-AI orchestrator and automate KPP reverse-flow and V&V traceability (Task 3).
- Integrate the scenario engine, ontology-maintenance pipeline, and digital-engineering deployment environment (Task 4).
- Deliver the Bayesian tranche-evolution engine, flagship demonstration, and transition package (Task 5).

Task / Technical Requirements

Task 1 — Formal Substrate (University of Arizona Lead)

Purpose: Task 1 establishes the formal substrate that anchors all downstream Phase II activities. It extends the Phase I mathematical, ontological, and morphism foundations into a complete formal basis for PWSA tranche-evolution reasoning.

Key Inputs: The contractor shall build on the Phase I STOIC ontology family, the initial morphism-based model-comparison framework, the Phase I DEVS-PWSA model family, and the preliminary digital-thread and agentic-AI concepts demonstrated in Phase I.

Scope & Activities: The contractor shall extend the STOIC ontology family to cover PWSA-specific concepts; shall instantiate the Homomorphism Library for representative PWSA subsystems; shall develop the Bayesian

morphism framework that governs belief propagation across tranche evolution; and shall specify the evidence-type provenance ontology required for epistemically grounded updates.

Key Outputs:

1. PWSA-extended STOIC ontology family
2. Homomorphism Library instantiation with structural and behavioral degradation measures
3. Bayesian morphism framework specification with partial-observation treatment
4. Evidence type provenance ontology aligned with PROV-O
5. Draft journal manuscript

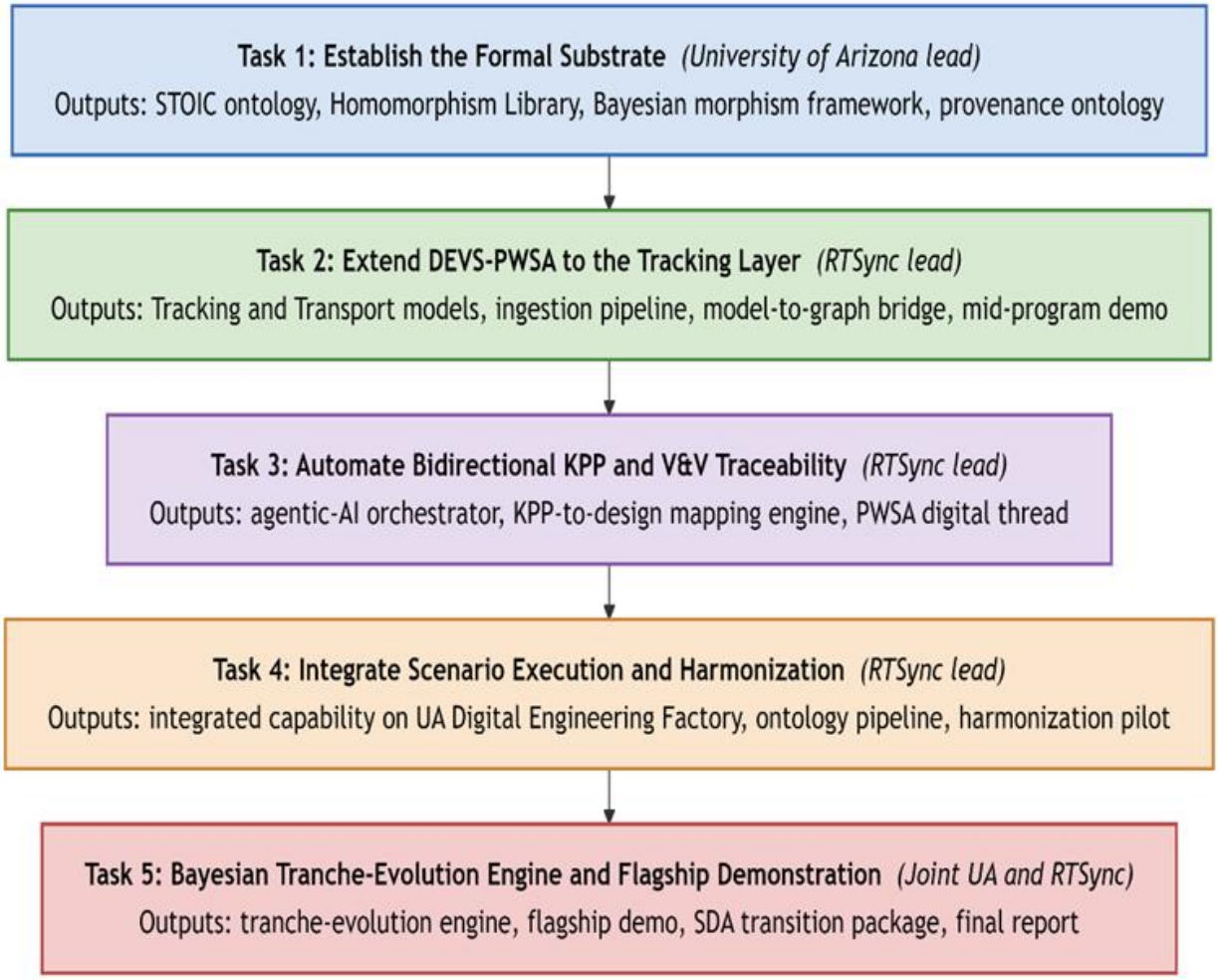


Figure 5. *Phase II Task Workflow to develop DEVS-PWSA Mission Engineering Engine for Tranche Evolution The workflow shown in Figure 5 presents this progression as a coherent, end-to-end pipeline.*

Task 2 — PWSA Scope Extension to Tracking (RTSync Lead)

Purpose: Task 2 extends the Phase I DEVS-PWSA modeling environment beyond the Transport Layer to include Tracking-Layer subsystems and establishes the ingestion backbone required for multi-tool digital-engineering integration.

Key Inputs: The contractor shall use the Task 1 formal substrate, the Phase I DEVS-PWSA Transport-Layer models, the Phase I ingestion-pipeline prototype, and the Phase I graph-database integration concept.

Scope & Activities: The contractor shall extend the DEVS-PWSA framework to represent Tracking-Layer behaviors including sensor pointing, custody handoff, and missile-warning event chains; shall implement the multi-tool digital-engineering ingestion pipeline; shall automate the model-to-graph-database bridge; and shall conduct the mid-program capability demonstration. The contractor shall align the ontology, model, and simulation tools with emerging open standards to make the pipeline self-sustaining as commercial tools evolve. The model-to-graph-database bridge synchronizes incrementally, re-validating only the subgraphs affected by a change at build and merge time rather than within the simulation loop. Ontology consistency and SHACL conformance therefore run over a small, slowly changing schema, while large-constellation execution remains the parallel DEVS engine's responsibility, preserving the rigor-at-speed posture as node counts scale into the thousands.

Key Outputs:

1. Extended DEVS-PWSA framework covering Tracking and Transport
2. Validated multi-tool ingestion pipeline
3. Automated bidirectional model ↔ graph-database bridge
4. Mid-program capability demonstration package

Task 3 — KPP Reverse-Flow and Automated V&V Traceability (RTSync Lead)

Purpose: Task 3 operationalizes automated design-to-KPP reasoning and V&V traceability, enabling continuous assessment of PWSA mission-thread performance and requirements coverage.

Key Inputs: The contractor shall use the Task 1 formal substrate, the Task 2 extended DEVS-PWSA models, the Phase I digital-thread methodology, and the Mission Thread Ontology.

Scope & Activities: The contractor shall implement the agentic-AI orchestrator baseline; shall automate bidirectional mapping between mission-level KPPs and subsystem-level design parameters; and shall apply the digital-thread methodology using the Mission Thread Ontology to surface traceability gaps, test coverage, and morphism-based degradation. The orchestrator operates under a propose-then-validate discipline: it coordinates deterministic tools and proposes model and mapping modifications, but no proposal is committed until it passes the two-tier validation gate established in Task 1 and operated in Task 4.2. Each proposed modification is checked for SHACL ten-shape conformance, dual-reasoner consistency, and competency-query coverage; proposals that fail any check are rejected before merge, so the agentic layer cannot alter the validated model family without satisfying the same machine-checkable constraints applied to human edits.

Key Outputs:

1. Agentic-AI orchestrator with tool adapters
2. Bidirectional KPP ↔ design-parameter mapping engine
3. PWSA digital-thread implementation with integrated decision-support view

Task 4 — Scenario Engine and Integration (RTSync Lead)

Purpose: Task 4 integrates the formal substrate, extended modeling environment, and automated reasoning components into a unified scenario-execution and digital-engineering capability deployable on the University of Arizona DEF.

Key Inputs: The contractor shall use the Task 1 formal substrate, the Task 2 extended DEVS-PWSA environment, the Task 3 orchestrator and traceability engine, and the Phase I geospatial digital-twin testbed.

Scope & Activities: The contractor shall integrate the scenario-execution and visualization layer; shall deploy the ontology-maintenance pipeline as a continuous-validation gate; shall host the integrated capability on the UA DEF; and shall conduct the cross-domain harmonization pilot.

Key Outputs:

1. Integrated DEVS-PWSA capability deployed on the UA Digital Engineering Factory
2. Scenario configurations for candidate Tranche 2 and Tranche 3 mission contexts
3. Operational ontology-maintenance and validation pipeline
4. Cross-domain harmonization pilot report and integrated demonstration package

Task 5 — Bayesian Tranche-Evolution Engine and Flagship Demonstration (Joint UA + RTSync Lead)

Purpose: Task 5 delivers the end-to-end tranche-evolution capability, demonstrating how PWSA mission-engineering confidence updates as new evidence, vendor models, and tranche scope changes arrive.

Key Inputs: The contractor shall use the Task 1 Bayesian morphism framework and provenance ontology, the Task 2 Tracking-Layer extensions, the Task 3 orchestrator and digital-thread implementation, and the Task 4 integrated scenario-execution environment.

Scope & Activities: The contractor shall operationalize the Bayesian tranche-evolution engine; shall apply the digital-thread methodology to tranche-evolution sensitivity analysis; shall execute the flagship Tranche-1-to-Tranche-2 demonstration; and shall produce the SDA transition package.

Key Outputs:

1. PWSA Bayesian tranche-evolution engine
2. Tranche-evolution sensitivity-analysis results
3. Flagship demonstration package
4. transition package and final technical report

Figure 6 summarizes the Phase II success definition as a vertical progression through Tasks 1–5. Each task box presents a brief description of the task's purpose and the corresponding success criterion that must be met for the task to be considered complete. Together, the tasks form a sequential workflow in which each stage builds on the validated outputs of the previous one, culminating in a fully integrated tranche evolution capability ready for evaluation for SDA.

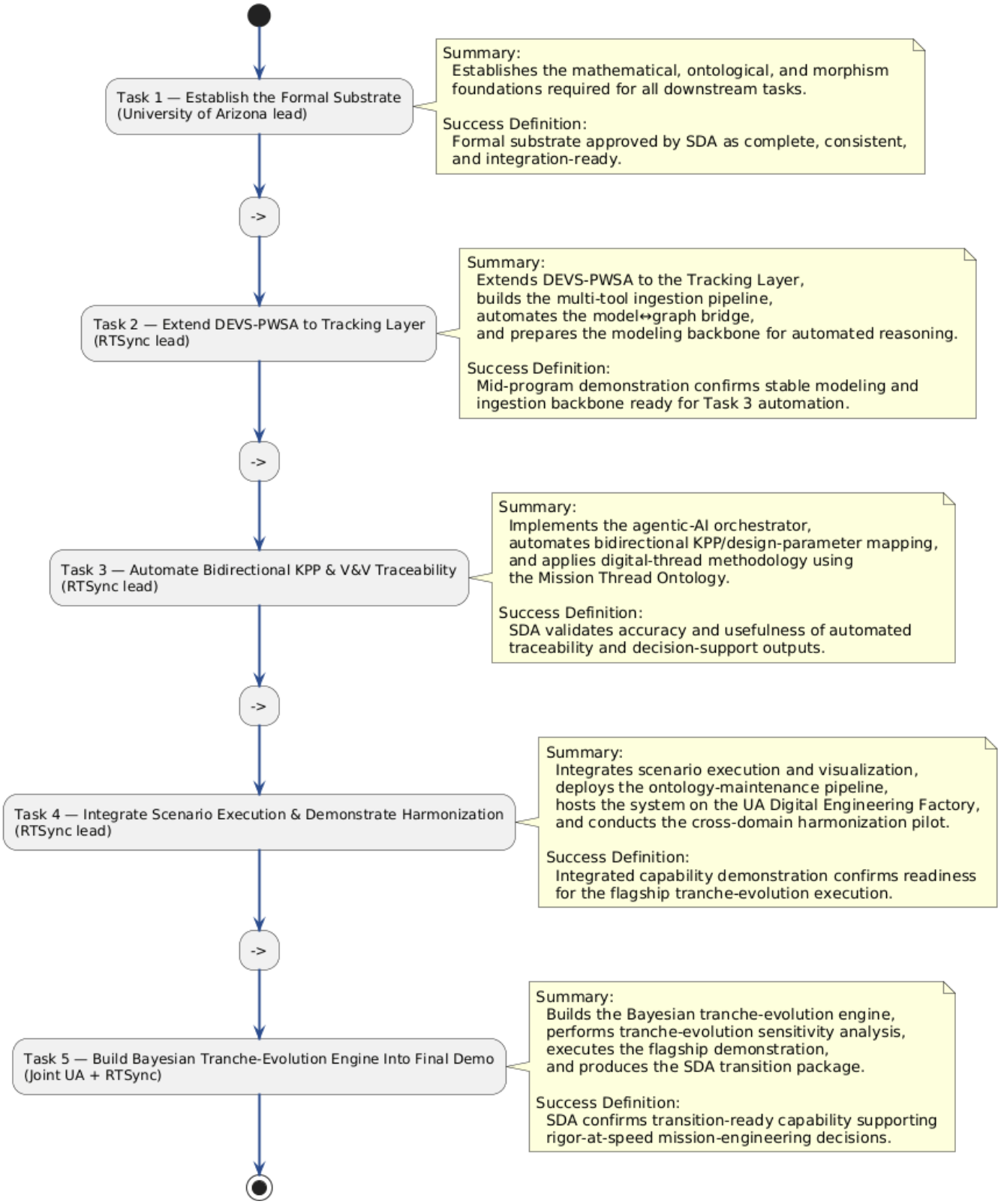


Figure 6. *Phase II Success Definition Across Tasks 1–5*

Timetable of Key Tasks and Activities

The contractor shall execute the 24-month period of performance across eight quarters, aligned to the milestones in Table 1. The schedule includes three major demonstrations (mid-program, integrated capability, flagship) and a final transition closeout. Each demonstration shall validate the integration of the preceding tasks as depicted in Figure 7.

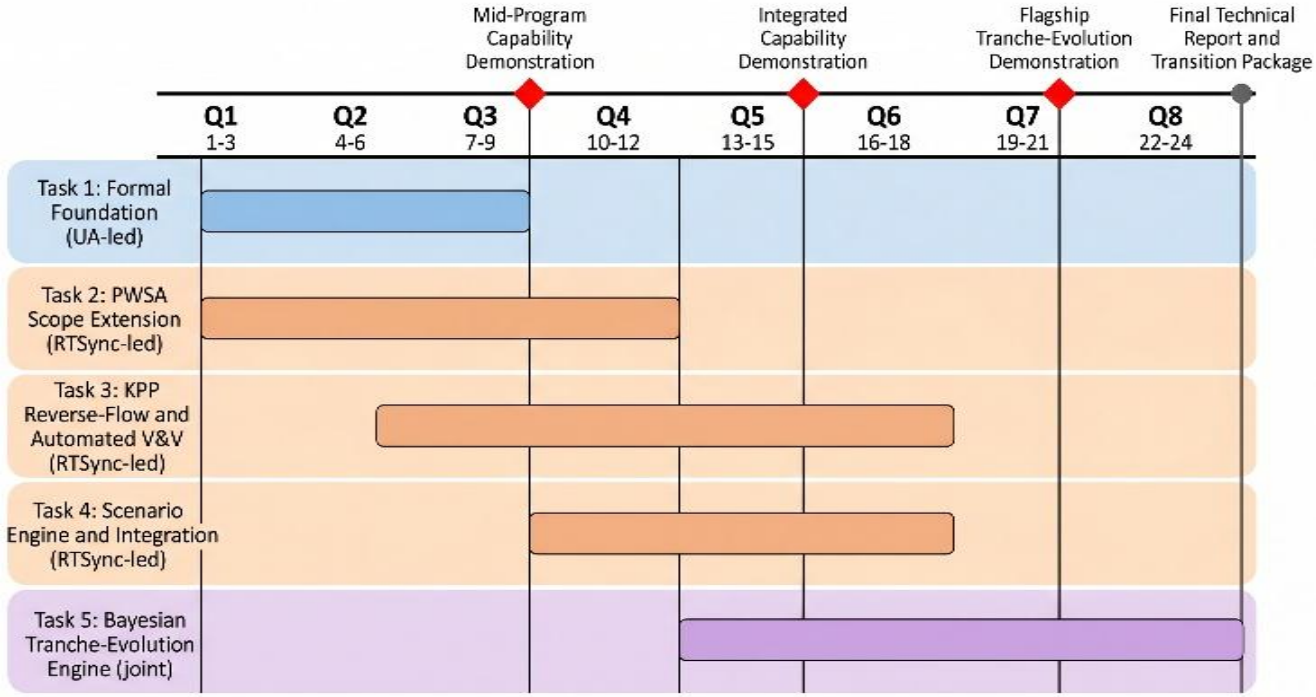


Figure 7. *Phase II Roadmap*

Five Tasks distributed across the 24-month period of performance, with three demonstration milestones at the end of Quarters 3, 5, and 7 (Months 9, 15, and 21) and final closeout at Quarter 8. The University of Arizona leads Task 1 (formal foundation); RTSync leads Tasks 2 through 4 (framework extension, automated V&V traceability, and integration); the two institutions jointly lead Task 5 (Bayesian tranche-evolution engine and flagship demonstration).

a. Timetable of Key Tasks and Activities

The 24-month period of performance is organized into eight quarters, with three demonstration milestones and a final closeout, as visualized in Figure 3. Quarter 1 (months 1-3): kickoff, requirements analysis, formal-foundation extension begins. Quarter 2 (months 4-6): Tracking-Layer extension begins, multi-tool ingestion pipeline stand-up, Homomorphism Library PWSA instantiation. Quarter 3 (months 7-9): mid-program capability demonstration of cross-resolution morphism quality measurement on Tracking and Transport. Quarter 4 (months 10-12): agentic-AI orchestrator baseline complete; evidence-type ontology integrated. Quarter 5 (months 13-15): integrated capability demonstration on the University of Arizona Digital Engineering Factory. Quarter 6 (months 16-18): KPP reverse-flow engine complete; cross-domain harmonization pilot complete. Quarter 7 (months 19-21): flagship PWSA tranche-evolution demonstration. Quarter 8 (months 22-24): final technical report; SDA transition package; software toolset hardened.

b. Risk Identification and Mitigation

Phase II risk management focuses on areas where execution exposure or external dependency could affect the schedule or scope. Table 2 identifies the principal risks, their relative probability and impact, and the team's planned mitigation. Risk status is reviewed at the mid-program demonstration (end of Quarter 3) and at the integrated capability demonstration (end of Quarter 5). Material risk updates are reported in the quarterly status reports and the monthly Technical Point of Contact briefings identified in the §7 Deliverables schedule.
How do the semantic validation gates (e.g., the SHACL graphs) programmatically override or constrain an autonomous agent's proposed model modifications?

Table 2. Principal Risks, Probability, Impact and Mitigation

Risk	**Probability**	**Impact**	**Mitigation**
External tool dependencies in the multi-tool ingestion pipeline (Task 2.2)	Medium	Medium	Modular adapter interfaces allow substitution; pipeline validated on two independently authored models; stable version targets locked at the kickoff meeting and reassessed quarterly.
Scope creep in cross-domain harmonization pilot (Task 4.4)	Medium	Medium	Pilot scoped to PWSA plus one partner domain; full multi-domain harmonization identified explicitly as a follow-on direction; quarterly scope review against the Phase II Roadmap baseline.
Agentic-AI orchestration standards evolution affecting Task 3.1	Medium	Medium	Lock to a specific protocol version at proposal acceptance; modular adapter pattern supports a single backend swap with bounded effort; reassessed at the mid-program demonstration.
Autonomous agent's proposed model modifications need semantic constraints	Medium	Medium	Semantic validation gates programmatically auto-correct to ensure that only formally valid artifacts progress to subsequent consideration [3]
Large network simulations cause severe lag in the real-time continuous-validation pipeline	Medium	Medium	The morphism-based simplification/abstraction chain illustrated in Figure 2 will be marshalled as needed to reduce execution times to acceptable limits.
Sponsor coordination on Task 2 kill-chain stretch scope	Medium	Low	Stretch scope explicitly conditional on Phase II Year 1 Technical Interchange Meeting concurrence; base scope unaffected if stretch is deferred.
Schedule pressure on the Quarter 7 flagship demonstration	Medium	High	Flagship demonstration baselined into Year 2; Quarter 8 reserved for hardening and transition package; integrated capability demonstration at Quarter 5 serves as a Quarter 7 rehearsal.
Multi-institutional coordination between the University of Arizona and RTSync	Low	Medium	Established collaboration with clear lead-and-support roles per Task as shown in Figure 5; weekly technical exchange call cadence; joint quarterly review aligned to the demonstration milestones.
External operational data availability for tranche-evolution validation	Low	Medium	Phase I scenario corpus extended with publicly releasable Tranche 1 and Tranche 2 mission contexts; controlled-access path through the University of Arizona Digital Engineering Factory if classified data required during Phase II.

The team's incident response process for material risk realization escalates within five business days through the cognizant Technical Point of Contact, with the principal investigators presenting a recovery plan no later than the following monthly briefing. The following deliverables correspond directly to the tasks outlined above and provide the measurable outputs of the Phase II effort.

(7) Deliverables

The RTSync-University of Arizona team shall provide all deliverables listed in the milestone table, including:

- Formal substrate artifacts (ontology family, Homomorphism Library instantiation, Bayesian morphism framework, provenance ontology).
- Extended DEVS-PWSA modeling environment and multi-tool ingestion pipeline.
- Agentic-AI orchestrator and automated KPP/design-parameter mapping engine.
- Integrated scenario-execution environment deployed on the UA Digital Engineering Factory.
- Bayesian tranche-evolution engine, flagship demonstration, and SDA transition package.

1. Review of Phase I Deliverables

Phase I delivered several core technical assets that establish the foundation for Phase II. These included the DEVS-PWSA framework as a family of models, a morphism-based application for mapping high-fidelity model parameters to lower-resolution models using rules based on an ontology in a description-logic profile covering both high- and low-resolution descriptions. The effort also produced graph-database instantiations of the ontology that was used to construct a simulation of a small satellite non-terrestrial network with two ground stations and a disruptor. Using a solar-flare experimental frame the simulation demonstrated the ability of the low-resolution model to support system redesigns that maintain reliable communications that would require much longer execution times in the high-resolution counterpart. Phase I also defined key follow-on priorities for Phase II. See the Phase I final report for details on proposed priorities, envisioned approaches, and technical rationale.

Phase II goals were further advanced in recent work. The RTSync-UA team participated in the 2026 Innovative Group for Next-Generation Integration, Testing, and Engineering (IGNITE) Hackathon. Hosted by the OSW(R&E) DTE&A (Office of the Under Secretary of Defense for Research and Engineering - Defense Test and Evaluation, and Assessments), with the TRMC (Test Resource Management Center), the event conducted an independent assessment of an authentic third-party mission-engineering model constructed from declassified mission documentation. A technical report on the team's output [33] outlines how the exercise successfully demonstrated the ability of multi-resolution family of models provide valid KPP measurements across seven defense-strength subsystems. It also demonstrated the end-to-end coupling of digital engineering artifacts to mission outcomes—a methodology that Phase II will scale to accommodate PWSA tranche evolution.

2. Transition from Phase I to Phase II

Phase I established the foundational elements required for Phase II execution. These include:

- A DEVS-based PWSA modeling family with validated stochastic extensions.
- An initial morphism-based model-comparison framework.
- A prototype digital-thread methodology for mapping requirements to test events.
- Early demonstrations of agentic-AI orchestration concepts.
- A preliminary ingestion pipeline and graph-database integration concept.

Phase II tasks are explicitly designed to extend, operationalize, and integrate these Phase I accomplishments. Each Phase II deliverable is anchored in a corresponding Phase I artifact, reducing technical risk and ensuring continuity across the development lifecycle.

The Phase II work plan to follow is structured to build directly on the technical foundations, prototypes, and validated methods delivered in Phase I and advanced in the IGNITE exercise. The formal DEVS-PWSA architecture, the morphism-based multi-resolution model-family framework, the early digital-thread demonstrations, and the agentic-AI orchestration concepts all form the foundation for Phase II work. Specifically, the ontology family will be broadened, and the morphism library will be instantiated for real PWSA subsystems. In addition, the prototype tranche development framework will be elaborated with automated ingestion and traceability pipelines, and the just described tranche-evolution Bayesian engine will be implemented.
Our planned products and reports to be delivered during Phase II include the products and reports in Table 1.

Phase II extends the Phase I DEVS-based mission-engineering framework for the PWSA along three convergent tracks: model-and-simulation with model-based systems engineering, Test and Evaluation, and digital engineering; agentic artificial intelligence; and ontology-based data and model harmonization. The scope progression carries Phase I Transport forward to Phase II Transport plus Tracking, with conditional stretch goals for kill-chain progression. The work is organized into five Tasks. Task 1 establishes the formal foundation. Task 2 extends the framework to the Tracking Layer. Task 3 implements automated Verification and Validation (V&V) traceability and the KPP reverse-flow committed in Phase I. Task 4 integrates the scenario-execution and visualization layer. Task 5 builds the Bayesian tranche-evolution engine and the flagship integrated demonstration. Figure 7 shows the Task structure across the 24-month period of performance, with three demonstration milestones at the end of Quarters 3, 5, and 7. The University of Arizona (UA) leads Task 1 and shares lead on Task 5; RTSync leads Tasks 2 through 4. The University of Arizona share of total effort targets approximately 31 percent.
Under Phase II, we will deliver a structured suite of technical reports and software products aligned with SDA's oversight requirements and the Contractor's Data Requirements list. These deliverables are designed to support SDA's mission of accelerating capability through modular, interoperable systems. Our reporting cadence includes seven quarterly status reports (.pdf) and 22 monthly technical briefings (.ppt) to the TPOC, ensuring transparency and alignment throughout the contract period. Technical documentation will include a Software Technical Design Document, User Documentation, Experimentation Results, and a Validation Results Report, each supporting SDA's emphasis on measurable progress and system-level integration.

Final deliverables will include the DEVS-PWSA software and source code (.zip), accompanied by comprehensive documentation (.pdf). At contract closeout, we will submit a Final Report with a non-proprietary summary suitable for DoD publication, along with all required SBIR/STTR lifecycle certifications, invention disclosures, software manuals, Phase III transition planning documents, and government property inventories. All software developed under this effort will be delivered; no hardware is planned for delivery. Should hardware prototyping occur, we will comply with DFARS 252.211-7003 for unique item identification and valuation.

(8) Related work

The RTSync–University of Arizona (UA) team brings a mature, continuously evolving technology base and a stable, experienced personnel line that directly underwrite the proposed DEVS-PWSA Phase II effort. RTSync contributes more than two decades of DEVS research, a production-ready simulation engine, and a record of government-funded transitions spanning mission engineering, kill-web analysis, and digital test and evaluation (dT&E).

UA's contribution builds on an established partnership with RTSync and the Department of Systems and Industrial Engineering, strengthened by the Arizona Center for Integrative Modeling and Simulation (ACIMS) heritage now resident at RTSync through Dr. Zeigler. Continuity of leadership is preserved: Dr. Wach—Co-PI for both SF24B-T009 and SF24C-T003 Phase I efforts—now leads the UA subaward as Researcher, carrying forward the Phase I ontology, DEVS-based PT&E framework, and LLM-assisted model-generation methods. These capabilities are augmented by UA's Digital Engineering Factory (DEF), Dr. Salado's NSF CAREER and SERC research portfolio, and UA's broader systems-engineering infrastructure.

Across prior efforts, the team has demonstrated the state-of-the-art integration of DEVS simulation, ontological modeling, MBSE, and exploratory analysis. Prior work has shown that combining DEVS with semantic technologies yields interoperable, reusable model families suitable for multi-performer environments. This aligns directly with emerging defense needs for interoperable, traceable, and composable modeling frameworks.

The team will also coordinate with the SDA Repository, leveraging high-fidelity models, mission-thread definitions, and Tracking Layer artifacts to refine the Phase II architecture and validate ingestion, ontology, and reasoning components. Repository access ensures that the prototype remains aligned with SDA's digital-engineering ecosystem and performer-supplied models.

RTSync's long history in SoS simulation, combined with UA's leadership in systems engineering and digital engineering, provides a robust foundation for integrating current and future SDA M&S tools, evaluations, and exercises. The Related Work portfolio confirms the team's distinctive expertise in model integration frameworks, DEVS-based composability, and automated V&V—capabilities that form the basis of the proposed DEVS-PWSA Phase II mission-engineering and tranche-evolution environment.

Title: **Cultivating PWSA Through Digital Life-Cycle Integration, Test, and Evaluation** (SF24B-T009 Phase I) Performing Organization/Client: RTSync (prime), Virginia Tech University (sub) / SDA Contract # (Value to date): FA254125PB002 ($179,992) Period of Performance: December 15, 2024 – March 2025 References: Mr. Gregory Grozdits, SDA greg.g.grozdits.civ@mail.mil (318) 436-9434 Mr. Tucker Mozeley, tucker.r.mozeley.civ@mail.mil 703-975-1754

Description: This Phase I effort directly supported SDA's mandate for speed, delivery, and agility by establishing the foundational digital-engineering, modeling, and phased test-and-evaluation (PT&E) methods required for future PWSA tranche evolution. The work focused on advancing SDA toward digital Test & Evaluation (dT&E) by developing mathematically grounded tools and processes that unify Model-Based Systems Engineering (MBSE), Modeling & Simulation (M&S), and mission-thread analysis within a coherent digital ecosystem.

Relevance to the Proposed Phase II Effort

The Phase I results form the direct technical foundation for the Phase II DEVS-PWSA mission-engineering engine. Phase II will scale and operationalize the Phase I prototypes by:

- Extending the ontology into a full formal substrate for tranche-evolution reasoning
- Expanding the modeling environment to the Tracking Layer and multi-performer ingestion
- Operationalizing automated KPP reverse-flow and V&V traceability
- Integrating all components into a unified scenario-execution environment
- Delivering a Bayesian tranche-evolution engine for continuous confidence updates

This continuity demonstrates a clear, credible, and low-risk transition path from Phase I research to a deployable, SDA-ready capability in Phase II.

Title: **DEVS Framework for Tranche-Based Mission Engineering and Performance Modeling of PWSA** (SF24C-T003 Phase I) Performing Organization/Client: RTSync (prime), Virginia Tech University (sub) /

SDA Contract # (Value to date): FA254125PB053 ($304,986) Period of Performance: August 1, 2025 – October 31, 2025, Reference: Mr. Richard Hencke, SDA richard.b.hencke.civ@mail.mil 703-624-7524

Description: The Phase I objective was to develop a prototype mission-engineering environment capable of representing evolving PWSA tranches and quantifying how architectural changes affect mission performance. The effort delivered several critical technical components that now serve as the substrate for the proposed Phase II DEVS-PWSA engine:

- A tranche-aware DEVS modeling family representing space-enabled behaviors, cross-domain interactions, and mission-thread dependencies.
- A formal parameter-mapping and model-family alignment method, enabling consistent transitions between models of varying fidelity and purpose (e.g., mission-level vs. subsystem-level).
- A preliminary morphism-based comparison framework to evaluate structural and behavioral differences between tranche configurations.
- A rigorous model-testing environment integrating RTSync's collaborative M&S platform with high-fidelity test and validation tool suite.
- Initial validation through a representative PWSA scenario, demonstrating the feasibility of DEVS-based mission engineering for tranche evolution.

The Phase I plan focused on four core activities, all of which were successfully executed:

- Define mission-engineering goals and performance criteria for tranche-based PWSA modeling.
- Develop and demonstrate algorithms for model-family alignment, including parameter mapping and cross-model consistency checks.
- Design and implement a rigorous model-testing environment suitable for multi-performer integration and iterative tranche updates.
- Validate the approach through a notional operational scenario, demonstrating applicability to real-world PWSA mission-engineering challenges.

Relevance to the Proposed Phase II Effort

The Phase I results provide the direct technical foundation for the Phase II DEVS-PWSA mission-engineering engine. Specifically:

- The DEVS model family becomes the core of the Phase II modeling environment.
- The parameter-mapping and morphism methods evolve into the Phase II Bayesian tranche-evolution engine.
- The model-testing environment becomes the basis for Phase II's scenario-execution and validation pipeline.
- The cross-domain mission-engineering constructs feed directly into Phase II's Tracking Layer expansion and multi-performer ingestion backbone.

Phase I demonstrated feasibility; Phase II will deliver a fully integrated, operationally relevant, transition-ready capability for SDA.

Title: **Paratemporal Simulation with Uncertainty Quantification** (MDA22-T001 Phase II) Performing Organization/Client: RTSync (prime), Binghamton University (sub) / MDA Contract # (Value to date): HQ086024C7600 ($1,509,994) Period of Performance: January 2024 – January 2026 Reference: Dr. Gregory Garbe, MDA gregory.garbe@mda.mil 719-721-9049

Description: RTSync developed ParaDEVS, an extension of the CAMLI modeling and simulation (M&S) environment to implement paratemporal system simulation techniques for application to execution of large stochastic models. RTSync implements techniques to achieve scalability, the ability to overcome the combinatorial explosion of branching arising from multiple choice points, and configurability, the ability to configure newly, or reconfigure conventionally, designed M&S artifacts to enable underlying simulators to support paratemporal execution. The development will leverage RTSync's support of collaborative development of DEVS models for reusability and composability, attributes that can overcome configurability

issues to support implementations that will foster wide-spread utilization in MDA's and similar defense and commercial M&S environments.

Relevance to the Proposed Phase II Effort

The ParaDEVS Phase II work provides a mature, validated, and high-performance simulation foundation that directly reduces technical risk for the DEVS-PWSA mission-engineering engine. Specifically:

- Thousand-fold speedups enable SDA to evaluate large PWSA constellations, multi-domain kill webs, and Tracking Layer behaviors at the pace required for tranche-based acquisition.
- Rare-event and resilience modeling directly supports SDA's need to assess degraded communications, cyber interference, node loss, and adversarial actions.
- Branching-time reasoning aligns naturally with tranche evolution, where each tranche introduces new payloads, waveforms, and vendor models that must be evaluated across multiple possible futures.
- Uncertainty quantification complements the Phase II Bayesian tranche-evolution engine, providing a mathematically grounded basis for evidence-weighted confidence updates.
- Configurability and composability inform the Phase II ingestion backbone, enabling rapid integration of heterogeneous models from SDA performers.

In short, ParaDEVS demonstrates RTSync's ability to deliver production-ready, mathematically rigorous, and massively scalable simulation technology—a decisive advantage for the DEVS-PWSA Phase II effort and a key enabler of SDA's rigor-at-speed mission posture.

Title: **Simulation Based Predictive Analytics on Autonomy and Artificial Intelligence Collaboration** (AF211-CSO1 Phase II) Performing Organization/Client: RTSync (prime), Embry Riddle Aeronautical University (sub) / Air Force Research Laboratory Contract # (Value to date): FA864922P1024 ($749,996) Period of Performance: June 2022 – August 2023 Reference: Dr. Nicholas Del Rio, nicholas.del_rio@us.af.mil

Description: For the Air Force Ventures (AFWERX) SBIR Phase II project, the RTSync team develops a Model Based System Engineering (MBSE) framework based on DEVS Modeling and Simulation (DMBSE) that supports predictive analysis to enhance design, development, testing, and evaluation of complex system of systems with mission threads and workflows. RTSync's simulation based predictive analytics platform will offer predictive capabilities to reduce development, integration, and deployment time and risk of new mission critical applications on autonomy and artificial intelligence (A2I) and lay the foundation for new business models. To achieve this objective, we aim to provide a prototype environment that employs our previously developed CAMLI to effectively construct, reuse, and adapt families of mission thread and work flow models to evolving requirements. We plan to demonstrate that the prototype DMBSE predictive analytics platform to enhance A2I for better mission-tailored decision support for researchers at Air Force Research Laboratories (AFRL) and other DoD, government, and commercial customers.

Relevance to the Proposed Phase II Effort

This project provides validated components directly supporting the proposed DEVS-PWSA engine:

- Mission-thread modeling + workflow orchestration → foundation for automated KPP reverse-flow and V&V traceability.
- Predictive analytics for autonomy → aligns with SDA's resilience and degraded-mode assessment needs.
- Reusable model-family structures → informs the Phase II ingestion backbone and multi-performer integration.
- AI-assisted model construction → complements Phase II's agentic-AI orchestration and digital thread automation.
- DEVS-MBSE integration → strengthens the formal substrate (ontology, morphisms, provenance, Bayesian reasoning).

Together, these results reduce technical risk and demonstrate RTSync's ability to deliver operational, DEVS-based predictive-analytics environments—directly enabling mission-engineering rigor at SDA's required pace

Title: **Model Level Integrated Simulation Architecture for Collaborative Development (**MDA18-012 2nd Phase II) Performing Organization/Client: RTSync (Prime), ORNL (sub), MDA Contract # (Value to date): HQ0860-22-C-7144 ($1,809,990) Period of Performance: September 2022 – September 2024 Reference: Dr. Gregory Garbe, MDA gregory.garbe@mda.mil 719-721-9049

Description: This Phase II effort developed RTSync's CAMLI-based Integrated Simulation Architecture, enabling development teams to integrate algorithmic models through a collaboration layer without modifying underlying simulators. The architecture provides rapid model integration, DEVS-based composability, high-performance execution, and streamlined data collection.

The work also advanced bidirectional SysML↔DEVS model mapping, expanded model-library and V&V metadata capabilities, and demonstrated how DEVS-based composability can represent kill-web structures already defined in SysML. Using the DEVS Streaming Framework for Adaptive Architectures (DSFA), the team instantiated alternative event pathways and orchestrated complex M&S compositions through experimental frames, supporting verification, validation, and system-performance evaluation.

Relevance to the Proposed Phase II Effort

This project provides key technical foundations for the DEVS-PWSA mission-engineering engine:

- Bidirectional SysML↔DEVS mapping directly supports Phase II's ingestion backbone and MBSE integration.
- CAMLI's collaborative integration layer aligns with Phase II's need to ingest heterogeneous performer models.
- DEVS-based composability and kill-web structuring map to PWSA mission-thread modeling and Tracking Layer behaviors.
- DSFA orchestration and experimental-frame execution inform Phase II's scenario-execution environment and automated V&V workflows.

Together, these results demonstrate RTSync's ability to deliver scalable, composable, and MBSE-aligned simulation architectures—reducing technical risk and accelerating transition for the DEVS-PWSA Phase II effort.

Title: **Automated Test Case Generator for Link 16** Performing Organization/Client: RTSync Northrop Grumman IT (Prime) /DITCO-SCOTT Joint Interoperability Test Command (JITC) Contract # (Value to date): NBC00H02C0001 ($4,663,041) Period of Performance: March 2008 – February 2013 Reference: Nancy Johnson, JITC 520-533-0421, nancy.johnson@disa.mil

Description: RTSync co-developed ATC-Gen, an automated standards-conformance testing system for Link-16 used by the Joint Interoperability Test Command (JITC). The effort formalized the entire MIL-STD-6016C rule set as a precise DEVS system model, enabling full analysis of message sequences, dependencies, and behavioral constraints. From this formal DEVS representation, ATC-Gen automatically generated mirror-image test cases that exercised the system under test and identified exact rule-set violations. The work demonstrated large-scale model formalization, automated test generation, and rigorous behavior-comparison workflows—capabilities that modernized JITC's testing processes and remain in operational use.

Relevance to the Proposed Phase II Effort

ATC-Gen provides proven foundations directly applicable to the DEVS-PWSA mission-engineering engine:

- Formal DEVS modeling of complex standards → supports Phase II's ontology, morphisms, and formal substrate.

- Automated test-case generation and behavior comparison → informs Phase II's automated V&V and KPP traceability workflows.
- Large-scale rule-set analysis and model consistency checking → aligns with Phase II's tranche-evolution reasoning and evidence-weighted validation.
- Operational transition success at JITC → demonstrates RTSync's ability to deliver deployable, mission-critical DEVS systems.

Together, these results strengthen the credibility and reduce the technical risk of the DEVS-PWSA Phase II effort.

Title: **Non-Real-Time Hardware Assisted Computer System Simulation (MDA20-T002 2nd Phase II)** Performing Organization/Client: RTSync (Sub), Oak Ridge National Laboratory (Sub) Contract # (Value to date): HQ086025CE0058 ($2,087,996) Period of Performance: September 2025 – September 2027 Reference: Dr. Greg Garbe, Missile Defense Agency, Gregory.Garbe@mda.mil 719-721-9049
Description: The RTSync team's design of a non-real-time simulation platform has created an opportunity to radically change the way that simulation utilizing enterprises such as Missile Defense Agency include high fidelity models of tactical software systems within their simulations. The new approach to developing and testing proposed System of System (SoS) architectures is to employ models of hardware systems that execute the unmodified operationally relevant software systems of interest, and thereby allow such software to be used in discrete event simulation models. The RTSync team implemented the sequential simulation protocol for supporting accurate non-real-time simulation of "as-is" operational software called DEVS-based virtual as is simulation test platform (DVAST). MDA plans to transition DVAST to become the standard method for testing the end-to-end performance of tactical software for the Golden Dome Initiative.
Relevance to the Proposed Phase II Effort
DVAST contributes directly to the DEVS-PWSA mission-engineering engine by providing:

- A proven method for integrating real operational software into DEVS models—supporting SDA's need to incorporate vendor-supplied behaviors and evolving tranche software.
- High-fidelity execution of complex SoS behaviors, aligning with PWSA's multi-domain mission-thread analysis.
- A validated DEVS-based platform for software-in-the-loop testing, informing Phase II's scenario-execution environment and automated V&V workflows.
- Scalable sequential-simulation protocols, strengthening Phase II's ability to evaluate large constellations and evolving tranche configurations.

Together, these results reduce technical risk and reinforce RTSync's ability to deliver operationally relevant DEVS-based mission-engineering capabilities for SDA.

Title: **Systems Engineering Beyond the Horizon (in contracting**) Performing Organization/Client: Stevens Institute of Technology/ Systems Engineering Research Center (prime), (sub) / Virginia Tech National Security Institute Contract # HQ0034-24-D-0023 (Value to date): ($1.1M) Period of Performance: September 2025 – September 2028 Reference: Mr. Kent Jones, kent.jones@nnsa.doe.gov;
Description: The digital transformation took many in the national security industry by surprise and has yet to reach it desired potential. The objective of this project is to seek disruption in order to get ahead of the technology and our adversaries. As such, this project aims to chart the future of systems engineering, create exemplars, and translate the research for implementations for the sponsor.
Relevance to the Proposed Phase II Effort
Systems Engineering Beyond the Horizon provides formal-foundation research directly applicable to the DEVS-PWSA mission-engineering engine:

- Charts the formal, math-based foundations of future systems engineering on which the Phase II formal substrate for tranche-evolution reasoning is built.

- Exemplar-driven translation of research into sponsor implementations mirrors the Phase II rigor-at-speed transition posture for the Space Development Agency (SDA).
- Provides direct continuity of the principal investigator's systems-engineering-foundations research line into the Phase II substrate.

Title: WRT 2406: **Optimized Portfolio Digital Engineering Transformation** (HQ003424D0023): Performing Organization/Client: Stevens Institute of Technology/ Systems Engineering Research Center (prime), (sub) / Virginia Tech National Security Institute Contract # HQ003424F0697 (Value to date): ($530,913) Period of Performance: September 2024 – September 2025 Reference: Mr. Kent Jones, kent.jones@nnsa.doe.gov;
Description: The defense industry is currently unable to maintain pace with the need; projects are consistently over cost, behind schedule, and have late realized risks; and the industry is in the midst of a transformation without understanding the interdependencies between technical (e.g., weapon) and social (e.g., organizational) systems. This research is characterizing optimized paths to achieve enterprise, portfolio, and program digital transformation; while exploring the cutting edge of the digital paradigm. Conway's Law and the Mirroring Hypothesis suggest that risks are mirrored between the socio- and technical- parts of an enterprise. This research tests the mirroring hypothesis, adds the unrecognized dimension of the digital engineering system, and asks the question of what are the sociotechnical barriers and levers to the digital transformation.
Relevance to the Proposed Phase II Effort
WRT-2406 provides digital-engineering transformation methods directly applicable to the DEVS-PWSA mission-engineering engine:

- Digital-engineering transformation methods inform the Phase II ingestion backbone and multi-performer integration.
- Portfolio- and program-level digital-engineering maturity assessment supports the transition-readiness narrative for SDA adoption.

Title: **Model-Based Mission Engineering** (N00174-22-1-0021): Performing Organization/Client: Virginia Tech National Security Institute Contract (Value to date): ($450,000) Period of Performance: June 2022 – June 2025 Reference: Dr. Caleb Strepka, caleb.r.strepka.civ@us.navy.mil;
Description: Mission engineering & analysis has become a critical function of DoD acquisition and system lifecycles as a mean to enhance mission success. This research explored advanced means of engineering and analyzing mission capabilities and needs. Methods leveraged largely focused on artificial intelligence (AI), such as machine learning (ML) used for network analysis and large language models (LLMs) used to expedite modeling of mission engineering artifacts.
Relevance to the Proposed Phase II Effort
Model-Based Mission Engineering provides foundations that feed directly into the DEVS-PWSA mission-engineering engine:

- Foundational to the Phase II mission-engineering engine; mission-capability analysis feeds the mission-thread and morphism-quality engine and the Tracking-Layer expansion.
- Artificial-intelligence and machine-learning network analysis, and large-language-model-assisted modeling of mission-engineering artifacts, feed the Phase II agentic-artificial-intelligence orchestration and digital-thread automation.

Title: **Tradespace Exploration for Better Verification Strategies**: Sponsor: Naval Postgraduate School. Value to date: $120,000. Period of Performance: 2017-2018. PI: Alejandro Salado.
Description: This research focused on improving verification strategies in large-scale systems, which are critical for demonstrating contractual fulfillment and reducing development risk. Current practices, guided by industry standards and expert judgment, face several risks, including uncertainty about the optimality of

verification strategies, lack of quantitative risk measurement, insufficient verification coverage, and misalignment with stakeholder objectives. To address these challenges, the research explored whether tradespace exploration can lead to more effective verification strategies. The project met two main objectives: (1) to develop a metric for assessing the value of verification activities based on the knowledge they reveal, and (2) to conduct a comparative analysis using a tradespace exploration tool. The findings were tested on a notional satellite system, with broader applications anticipated for defense, space, automotive, and other systems. The outcomes aimed to enhance the cost-effectiveness and value of verification strategies, with potential public benefits in early safety and efficacy of commercial products and services.

Relevance to the Proposed Phase II Effort

This research provides verification-value foundations directly applicable to the DEVS-PWSA mission-engineering engine:

- The metric for the value of verification activities underpins the Phase II evidence-weighted verification-and-validation (V&V) and key-performance-parameter (KPP) traceability workflows.
- Tradespace comparison methods inform structured comparison of tranche configurations.

Title: **Dynamic Contracting of Verification Activities by Applying Set-based Design to the Definition of Verification Strategies**: Sponsor: Naval Postgraduate School. Value to date: $119,000. Period of Performance: 2018-2019. PI: Alejandro Salado.

Description: This research sought to improve the definition of verification strategies in large-scale systems, addressing the limitations of existing practices where fixed verification strategies are agreed upon at the start of acquisition programs. Since these strategies are set when system knowledge is minimal, they often become suboptimal as system development progresses, leading to inefficiencies and missed opportunities to optimize verification activities. To tackle this, the research explored whether set-based design can enable dynamic contracting of verification activities, allowing for more adaptive and valuable strategies. The research objectives included generating optimal future verification paths and comparing set-based design against current practices. This approach was tested on a notional satellite system and showed that it can enhance the cost-effectiveness and value of verification strategies across various sectors, such as defense, space, automotive, and public infrastructure.

Relevance to the Proposed Phase II Effort

This research provides adaptive-verification methods directly applicable to the DEVS-PWSA mission-engineering engine:

- Adaptive, knowledge-dependent verification aligns with tranche evolution, where verification strategies must update as each tranche adds payloads, waveforms, and vendor models.
- Set-based generation of future verification paths informs the Bayesian tranche-evolution engine's continuous-confidence updates.

Title: **Collaborative Research: Maximizing the Expected Value of Engineered Systems Through Coordinated Verification**: Sponsor: National Science Foundation. Value to date: $298,625. Period of Performance: July, 2018 through July, 2021. PI: Alejandro Salado.

Description: The objective of this research was to mathematically derive value-maximizing system verification strategies in multi-firm engineering projects, and to identify contractual agreements between firms that can implement these strategies. Verification is a crucial activity in the engineering of large-scale systems, but identifying superior verification strategies is challenging. Firms need to decide when to execute verification activities, at which system integration level, and with which accuracy or confidence. In their decisions, each firm needs to consider how the verification activities executed by other firms affect the effectiveness of their own verification strategy. From a system customer/user perspective, the aggregated verification strategies of all firms at all system integration levels should contribute to maximizing the value of the overall system. However, firms seek to optimize their individual objectives instead. Hence, this conflict of interest between customer and firms, and between firms, needs to be overcome. Well-designed contracts and incentive

mechanisms can align the interests of all parties when defining verification strategies. We developed a mathematical framework to address these challenges and build a scientific theory of verification.

Relevance to the Proposed Phase II Effort

This research provides multi-firm verification foundations directly applicable to the DEVS-PWSA mission-engineering engine:

- Multi-firm, value-maximizing verification coordination maps to the Phase II multi-performer ingestion backbone and to aggregating evidence across heterogeneous SDA performers.
- The mathematical framework for a theory of verification supports the Phase II formal evidence substrate.

Title: **CAREER: Belief Formation in Verification Strategies and the Evaluation of Verification Evidence**: Sponsor: National Science Foundation. Value to date: $500,000. Period of Performance: September, 2019 through September, 2025. PI: Alejandro Salado.

Description: This research pioneers the investigation of how engineers process verification expectations and evidence at the nexus of cognitive models and Bayesian analysis. Capturing how engineers transform objective data into subjective beliefs is the cornerstone of this program. Modeling the information structures by which engineers build confidence in a system's performance and behavior based on past and planned verification activities can markedly enhance the scientific understanding of verification in SE. This research combines Bayesian analyses, traditional and dynamic belief graphs (with precise and imprecise probabilities), and decision theory with empirical observations of professional engineers and information artifacts (e.g., verification plans and reports) to establish explanatory models of how engineers' beliefs form. This project goes beyond identifying theoretical insights by contextualizing how these insights fit within actual SE practice.

Relevance to the Proposed Phase II Effort

This research provides the Bayesian evidence foundations central to the DEVS-PWSA mission-engineering engine:

- Direct mathematical foundation for the Phase II evidence-type ontology (established, inferred, synthetic) and the Bayesian tranche-evolution engine.
- Formalizes how engineer confidence forms from verification evidence using Bayesian analysis and belief graphs, the core of evidence-weighted tranche assessment.

Title: **WRT-1071: Digital Transformation in Test and Evaluation for AI/ML, Autonomous, and Continuously Evolving Systems**: Sponsor: Stevens Institute of Technology / Systems Engineering Research Center. Value to date: $420,000. Period of Performance: September 30, 2022 through September 30, 2024. PI: Alejandro Salado.

Description: This project utilized Model-Based Systems Engineering (MBSE) and ontologies to establish

Relevance to the Proposed Phase II Effort

WRT-1071 provides ontology-based test-and-evaluation foundations directly applicable to the DEVS-PWSA mission-engineering engine:

- The ontology-based Test and Evaluation Master Plan informs the Phase II automated V&V traceability and KPP reverse-flow.
- Integrating test-and-evaluation requirements and processes into system models supports the evidence substrate and digital-thread automation for continuously evolving tranches.

a digital engineering approach for test and evaluation (T&E) planning. By integrating T&E requirements and processes into system models, the project enabled dynamic, data-driven updates to test plans throughout the system lifecycle. Ontologies were used to formalize key concepts and relationships, creating an ontology-based Test and Evaluation Master Plan (TEMP) that linked T&E data with other engineering models for continuous refinement. This approach improved traceability, automated decision-making, and system

integration, offering a more efficient and adaptable T&E process for complex engineering domains like defense and aerospace.

Title: **WRT-1095: Transforming Systems Engineering through Digital Engineering**: Sponsor: Stevens Institute of Technology / Systems Engineering Research Center. Value to date: $399,700. Period of Performance: January 1, 2025 through December 31, 2025. PI: Alejandro Salado.
Description: The project utilizes MBSE and ontologies to develop novel digital engineering capabilities and workflows. A particular focus is put towards generating dedicated dashboards that allow to consume data across a suite of tools without depending on the source tools, as well as integrating mathematical optimization techniques into the design and execution of test and evaluation strategies in a digital engineering environment.
Relevance to the Proposed Phase II Effort
WRT-1095 provides digital-engineering and optimization methods directly applicable to the DEVS-PWSA mission-engineering engine:

- Tool-agnostic consumption of data across a tool suite informs the Phase II heterogeneous performer-model ingestion backbone.
- Mathematical optimization within test-and-evaluation strategy execution supports the Phase II scenario-execution and validation pipeline.

Title: **MBSE CoPilot for Small Projects**: Sponsor: Jet Propulsion Laboratory/NASA. Value to date: $40,000. Period of Performance: October 15, 2024 through October 14, 2025. PI: Alejandro Salado.
Description: The project aims to lower the barrier to using Model-Based Systems Engineering (MBSE) tools significantly by incorporating an Artificial Intelligence (AI)-based natural language interface (NLI) as an alternative to traditional user and programmatic interfaces (UI and API). This NLI, referred to as CoPilot, would enable users to interact with an MBSE tool through text prompts to help author and analyze models. This innovation is expected to reduce the need for systems engineers to pre-learn the intricacies of the MBSE language, methodology and UI of the tool. The primary objectives are to simplify the processes of creating system models, analyzing system models, and generating required products from system models. The feasibility of these capabilities will be validated in the context of a small flight project.
Relevance to the Proposed Phase II Effort
MBSE CoPilot provides AI-assisted modeling methods directly applicable to the DEVS-PWSA mission-engineering engine:

- The artificial-intelligence natural-language interface for Model-Based Systems Engineering authoring complements the Phase II agentic-artificial-intelligence orchestration and language-model-assisted model generation.
- Lowering the barrier to model authoring supports rapid multi-performer model onboarding in the ingestion backbone.

Building on this foundation of prior research and operational prototypes, the Phase II system also offers clear commercialization pathways

(9) Commercialization Potential

The DEVS-PWSA platform is purpose-built to meet the Space Force's growing need for agile, data-centric, and mathematically grounded mission-engineering tools that can keep pace with the rapid evolution of the Proliferated Warfighter Space Architecture (PWSA). By integrating DEVS formalism, MBSE alignment, and agentic-AI orchestration, DEVS-PWSA enables continuous tranche-evolution analysis—allowing SDA to rapidly assess how new payloads, waveforms, and vendor updates affect mission performance across successive tranches.

RTSync, in collaboration with the University of Arizona, is well-positioned to transition this capability from research to operational use. The platform's ingestion backbone, automated traceability, and Bayesian update engine directly support SDA's requirement for rigor-at-speed mission engineering, enabling rapid integration of heterogeneous models and evidence-based evaluation of evolving architectures.

Although optimized for PWSA, the underlying DEVS-based simulation architecture has broad commercial and defense applicability. The same framework can model autonomous-vehicle interactions, optimize air-traffic-management operations, support emergency-response planning across large geographic regions, and evaluate healthcare or logistics strategies under uncertainty. These cross-sector applications demonstrate strong dual-use potential and expand the addressable market beyond space defense.

RTSync's commercialization approach begins with targeted deployment in space-defense and financial-analytics markets, where DEVS-PWSA's multi-fidelity modeling and tranche-evolution capabilities provide immediate value. Subsequent phases will extend into adjacent sectors—autonomous systems, air-traffic management, emergency response, and healthcare digital twins—where the platform's mathematical rigor and agentic-AI reasoning offer clear competitive advantages.

Through strategic partnerships, proven transition pathways, and cross-sector applicability, RTSync is positioned to commercialize DEVS-PWSA as a scalable, interoperable, and high-impact solution for mission engineering, tranche evolution, and digital-engineering decision support across both defense and commercial markets.

1. Commercialization Plan

a. First Planned Product Incorporating the Proposed Technology

The first commercial product will be the DEVS-PWSA Mission Engineering Engine, delivered as a containerized software suite with ingestion adapters, Bayesian tranche-evolution engine, automated KPP traceability module, and scenario-execution environment. This product will be deployable within the Space Development Agency (SDA) and other DoD digital engineering environments.

b. Probable Customers and Estimated Market Size

The DEVS-PWSA Mission Engineering Engine addresses a rapidly expanding market driven by DoD digital-engineering mandates, proliferated LEO architectures, and the need for high-fidelity mission-simulation environments. The digital engineering and mission-simulation markets exceed $3.5B annually, with approximately $800M attributable specifically to DoD space-mission analysis. Broader industry trends reinforce this opportunity: the global simulation-software and services market was valued at $16.55B in 2022 and is projected to reach $40.5B by 2030 at an 11.83% CAGR. A separate MarketsandMarkets report projects the simulation-software market to reach $26.9B by 2026, underscoring sustained demand for advanced modeling and simulation capabilities.

Within this landscape, the DEVS-PWSA toolkit occupies a high-value niche. Advanced kill-web simulation, multi-fidelity ingestion, and mathematically grounded tranche-evolution reasoning represent a small but strategically critical segment of the overall market. By specializing in these capabilities—rather than offering generalized simulation tools—RTSync differentiates itself from competitors and targets customers who require advanced, evidence-driven mission-engineering frameworks.

Primary DoD customers include: Space Development Agency (SDA), U.S. Space Force, Space Systems Command (SSC), Missile Defense Agency (MDA), Air Force Research Laboratory (AFRL/RV), PEO Space,

PEO C3I, PEO Missiles & Space. These organizations require continuous architecture assessment, model ingestion, and mission-thread analysis for proliferated LEO constellations, missile-warning architectures, and multi-domain kill webs.

Commercial customers include major aerospace primes and constellation operators such as: Northrop Grumman, Raytheon, Lockheed Martin, L3Harris, Amazon Kuiper, SpaceX Starshield, Blue Origin, and Sierra

Space. These companies face increasing demand for digital-engineering toolchains that support rapid model integration, scenario execution, and mission-performance evaluation.

By focusing on advanced kill-web simulation, DEVS-based stochastic modeling, and automated tranche-evolution reasoning, RTSync is positioned to capture a distinct, defensible market segment within the broader simulation-software ecosystem. This specialization enables RTSync to serve customers who require high-fidelity, mathematically grounded mission-engineering capabilities that general-purpose simulation tools cannot provide.

The following Table 3 provides RTSync's rough estimates of initial commercial potential in the envisioned DEVS-PWSA application areas:

Table 3 *Market Size*

Market Name	Known or Estimated Customer Organizations	% Using or Needing Similar/Related Capabilities	Potential Market Size	Annual Budget for Similar or Related SW per Customer	Total Expenditure	Initial ~1% Penetration
Defense	30	100%	30	$20M	$600M	$6M
Engineering	50	100%	50	$4M	$200M	$2M
Cybersecurity	30	50%	15	$2133K	$32M	$320K
Manufacturing	123	30%	37	$1M	$37M	$370K

RTSync estimates an initial **niche market** of $869M across defense, engineering, cybersecurity, and manufacturing sectors, with a **1% penetration goal** yielding $8.7M in early revenue potential.

c. Funding Requirements and Strategy

In projecting financial returns and investment needs beyond Phase I and II of this work, Table 4 presents a five-year forecast based on capital and cash flow requirements necessary to achieve revenue targets to support ongoing development and expansion.

Table 4. *Five-year financial forecast*

	Year 1	Year 2	Year 3	Year 4	Year 5
Revenues	$250,000	$550,000	$1,200,000	$2,500,000	$3,700,000
Costs	$550,000	$400,000	$300,000	$250,000	$250,000
Profit	($300,000)	$150,000	$900,000	$2,250,000	$3,450,000
Investment	$400,000	$250,000	0	0	0

RTSync estimates that $2.5M in additional development, integration, and certification funding is required to reach TRL-7 and full operational deployment. This funding will be secured through:

- Phase III contracts with SDA for Tranche 2 and Tranche 3 architecture-analysis lines
- SSC Digital Engineering Enterprise integration funding
- Commercial licensing revenue from aerospace primes and constellation operators
- Strategic investment from industry partners seeking DEVS-based digital-engineering capabilities

This diversified funding approach ensures a stable transition path beyond Phase II.

d. Marketing Expertise and Strategy

RTSync has strong technical and transition experience but is expanding its dedicated marketing capacity as it prepares to commercialize the DEVS-PWSA Mission Engineering Engine. During Phase II, RTSync will hire personnel with expertise in digital marketing, analytics, and defense-sector outreach to strengthen visibility and customer engagement.

To accelerate commercialization, RTSync will complement internal hiring with targeted external efforts, including partnerships with established defense contractors, engagement with DoD innovation hubs such as AFWERX, DIU, and SpaceWERX, and focused outreach to SDA, SSC, MDA, and AFRL program offices. RTSync will also participate in key defense-industry conferences and SBIR transition events to showcase the DEVS-PWSA toolkit.

This combined approach ensures RTSync will have the marketing capability needed to successfully transition the technology to both DoD and commercial space-engineering markets.

e. Competitors and Anticipated Price/Quality Advantage

RTSync operates in a competitive market, with key players in aerospace, defense, and financial analytics including Lockheed Martin, Northrop Grumman, and Bloomberg.

Competitors include AGI/Ansys STK, AFSIM-based integrators, LinQuest, Parsons, BlueHalo, and Raytheon digital-engineering toolchains. RTSync's competitive advantages include:

- 70–90% lower compute cost due to CPU-only execution (no GPU training pipelines)
- 50–80% faster analysis cycles through automated KPP reverse-flow and traceability
- Hours-not-weeks model ingestion via the multi-tool ingestion backbone
- Mathematically grounded reasoning (morphisms + Bayesian updates) not available in competitor tools
- Full transparency and auditability, unlike opaque ML-based approaches

These advantages position RTSync as a cost-effective, high-fidelity alternative to existing mission-analysis platforms.

By capitalizing on AI-driven MBSE and simulation advancements, strategic partnerships, and a diversified revenue model for DEVS-PWSA technology, RTSync is poised to lead the next generation of predictive financial modeling and space defense simulation solutions for PWSA. RTSync is aggressively pushing the boundaries with a bold, ambitious vision for the next generation of M&S software, redefining how advanced simulations should be developed, deployed, and leveraged for strategic use.

Intellectual Property (IP)

RTSync's proposed Phase II project aims to overcome key technical challenges in developing generic DEVS-PWSA technologies. As part of its intellectual property strategy, RTSync plans to file a new patent to safeguard DEVS-PWSA's innovations by November, 2026, complementing its existing patent applications. From its earlier work, RTSync was granted the necessary patents. The company's IP strategy includes: 1) Patents: Securing IP protection for core technologies in 2023 and 2024, including: a) US Patent 12,124,778 B1: "Method for Improving the Simulation of Complex Stochastic Systems." b) US Patent 11,922,102 B1: "Method to Integrate Diverse Components for Simulation of Complex Systems." 2) Trade Secrets: Protecting proprietary AI-driven modeling techniques and paratemporal simulation algorithms. 3) Publications & Thought Leadership: Establishing credibility through peer-reviewed research, including contributions to the Winter Simulation Conference. This multi-layered strategy ensures sustained innovation, market differentiation, and legal protection against industry threats.

2. Commercialization Strategy Plan

Through proven product and program partnering processes, RTSync will commercialize the proposed DEVS for PWSA technology for government and private sector customers. RTSync has a proven track record of M&S technology transfer from theoretical innovation to real-world practice. Founded in 2004, RTSync has developed and applied DEVS technology and software to interoperability testing and evaluation, and deployment of this DoD award-winning technology.

a. Private Sector or Non-SBIR/STTR Governmental Funding Sources

The DEVS-PWSA platform has already attracted significant interest from both government and commercial organizations outside the SBIR program, demonstrating strong commercialization potential. RTSync's collaboration with the Missile Defense Agency's Modeling & Simulation Directorate (DES)—the steward of the CAMLI framework—provides a direct, non-SBIR pathway for integrating DEVS-PWSA into broader DoD digital-engineering environments. This complements SDA's near-term needs for PWSA Tranche 1 and Tranche 2 mission-analysis support, including regional persistence, missile-warning assessment, and beyond-line-of-sight targeting analysis.

Commercial and defense-adjacent markets are also showing early commitment. Satellite-constellation operators, aerospace primes, and emerging space-services companies face challenges similar to SDA's, such as rapid model integration and continuous architecture evolution. The DEVS-PWSA ingestion backbone, automated traceability, and Bayesian evolution engine directly address these needs, offering a vendor-agnostic foundation for digital engineering ecosystems in commercial space, autonomous-systems certification, and safety-critical digital twins.

RTSync's partnership with other defense contractors including Northrop Grumman further strengthens its commercialization posture by providing a channel for integrating DEVS-based capabilities into major programs such as SDA's LEO PWSA, the Army's IBCS, STSS, and SBIRS. These programs represent substantial non-SBIR funding opportunities for mission-thread analysis and digital test and evaluation.

Most importantly, RTSync has already secured over $800,000 in non-SBIR/STTR revenue-generating projects that leverage the same foundational technologies as DEVS-PWSA. These include healthcare digital twins, financial-system analytics, automated scenario generation, and Army campaign-model modernization. Table 5 summarizes some of the commitments and demonstrates clear market traction and technical relevance across diverse sectors.

Table 5. Non-SBIR/STTR potential revenue-generating projects

Project Title	Customer	Schedule	Amount ($)
Simulation based Continuity of Care (SimCo) and Digital Twin	Johns Hopkins University Kennedy Krieger Institute (KKI)	October 2025 – July 2026	184,000
DEVS and MBSE Support for Campaign Model Modernization Program by US Army CAA	Trideum (prime)	December 2025 – April 2027	650,000

b. Phase III follow-on commitments for the research subject

RTSync's DEVS-based Virtual As-Is Simulation Test platform (DVAST) under MDA20-T002 topic provides a proven and credible pathway for Phase III transition, particularly with the Missile Defense Agency. DVAST has already received substantial STTR investment, including a $1.51M Phase II award (2022–2024) and a second Phase II award totaling $2.09M (2025–2027), as shown in Table 6. These continued investments demonstrate MDA's sustained commitment to RTSync's DEVS-based simulation and MBSE technologies and their applicability to digital-only test venues, hardware-in-the-loop validation, and tactical software integration. Recent discussions with MDA senior leadership indicate plans to allocate an additional $5 million in development funding to expand the DVAST 2nd Phase II enhancement effort, further reinforcing the agency's long-term commitment to RTSync's DEVS-based digital-engineering capabilities and the transition to Golden Dome Initiative program.

The proposed DEVS-PWSA platform is architected to extend DVAST's capabilities into broader mission-engineering, kill-web analysis, and tranche-based performance modeling. Because DEVS-PWSA

shares core formalisms, ingestion pipelines, and simulation protocols with DVAST, it is technically aligned for rapid integration into DVAST's operational workflows. RTSync anticipates Phase III follow-on funding to incorporate DEVS-PWSA into MDA and SDA mission-analysis environments, enabling enhanced scenario generation, performance evaluation, and evidence-based decision support across evolving PWSA tranches.

This alignment ensures that DEVS-PWSA is not only compatible with existing MDA digital-engineering infrastructure but also strategically positioned for rapid deployment. The shared architecture, demonstrated performance, and ongoing MDA investment collectively provide a strong foundation for Phase III transition and long-term operational adoption.

Table 6. DVAST - potential Phase III transition project for MDA

Project Title	Customer	Schedule	Amount ($)
DEVS-based virtual as is simulation testing platform (DVAST)	Missile Defense Agency STTR Phase II HQ086022C7601	June, 2022 – June 2024	1,509,996
DEVS-based virtual as is simulation testing platform (DVAST)	Missile Defense Agency STTR Second Phase II HQ086025CE058	September, 2025 – September 2027	2,089,992

c. Other Indicators of Commercial Technology Potential

RTSync's commercialization strategy for the DEVS-PWSA platform is grounded in demonstrated transition success, clear market alignment, and a scalable business model. The firm has repeatedly shown its ability to move DEVS-based modeling and simulation technologies from research prototypes into operational use across both defense and commercial sectors. The DEVS-PWSA solution, integrating DEVS, MBSE, and agentic AI extends this track record by offering a modular, interoperable, and scalable toolset directly aligned with SDA's tranche-based development model and broader DoD digital-engineering initiatives.

RTSync's commercialization strategy includes:

- **Targeted Market Entry:** Initial deployment within SDA, MDA, and U.S. Space Force programs, followed by expansion into adjacent markets such as autonomous systems, air-traffic management, and financial analytics where DEVS-based reasoning and multi-fidelity ingestion provide unique value.
- **Diversified Revenue Model:** Leveraging SBIR Phase III pathways, strategic partnerships with prime contractors (e.g., Northrop Grumman), and selective private investment to support productization, sustainment, and scale.
- **Financial Growth Plan:** A five-year forecast projecting profitability by Year 3 and revenues reaching **$3.7M by Year 5**, supported by a planned **$650K capital raise** following Phase II to accelerate commercialization and customer onboarding.
- **Intellectual Property Protection:** Existing patents and planned filings secure RTSync's innovations in stochastic modeling, simulation architecture, and AI-driven decision support, strengthening competitive defensibility.
- **Cross-Sector Applicability:** The mathematical rigor and agentic AI capabilities of DEVS-PWSA enable adoption in healthcare digital twins, emergency-response planning, manufacturing optimization, and other high-assurance domains—expanding commercial reach beyond defense.

These indicators, combined with RTSync's active Phase II contracts, the DVAST Phase III transition pathway, and growing non-SBIR revenue demonstrate strong commercial potential and a credible strategy for exploiting it. RTSync is well-positioned to commercialize DEVS-PWSA as a transformative capability for digital engineering, mission analysis, and evidence-based decision support across defense and commercial markets.

(10) Relationship with Future R/R&D Efforts

1. Anticipated results of the proposed approach

The Phase II period will serve as a catalyst to commercialization of the DEVS-PWSA for potential applications throughout DoD. At successful completion of Phase II, the RTSync Team anticipates the following results:

1. Prototype implementation of proposed DEVS-PWSA mission engineering engine
2. A demonstration of the design concepts in a software execution showing how to apply them in a realistic simulation designed for PWSA Tranche Evolution of interest to SDA.

The anticipated result during second Phase II period is that the DEVS-based PWSA mission engineering engine that will demonstrate a scenario that reflects the complexity of real-world PWSA tranche evolution allowing us to test the approach's scalability and adaptability. After Phase II, the SDA customer could support a transition with a Phase III program to enhance our DEVS-PWSA environment by testing various use cases and performance analysis applications for robust operational capabilities.

The RTSync team does not anticipate certifications or approvals are required to conduct Phase III testing for this proposed approach. If classified data access is required for Phase III activities, the RTSync team will identify the need with SDA and prepare for necessary facility clearance form and process during Phase II period. The DEVS-PWSA platform can be leveraged to enhance the Digital Engineering ecosystem infrastructure across Department of Defense (DoD) agencies. The RTSync team will engage with defense contractors and major industry players to identify potential use cases and explore opportunities for future commercialization of DEVS-PWSA technologies.

Potential follow-on Phase III will build on the methodological foundation established in Phase II. The Phase II project extends the prototype implemented in DEVS-PWSA to achieve significant progress toward an industrial strength tool set. Analysis of achievement criteria with the AFRL TRL Calculator suggests that Phase II will conclude at TRL 7 or 8 level, demonstrating working prototype.

2. Phase III Transition and Operational Adoption

The Phase II effort positions the PWSA mission-engineering capability for Phase III transition along three convergent pathways aligned with the Space Development Agency's 2026 rigor-at-speed and vendor on-orbit acceptance posture.

The first pathway transitions the integrated capability to SDA Mission Engineering and Test and Evaluation offices as a working environment for tranche-evolution assessment. The Quarter 8 transition package includes a deployment guide for the University of Arizona Digital Engineering Factory and equivalent on-premises environments, user documentation for the agentic orchestrator and the Bayesian tranche-evolution engine, ontology family documentation with examples, and a reference implementation of the multi-tool ingestion pipeline. The transition path explicitly supports the new SDA posture in which vendors are required to conduct on-orbit testing before SDA formally accepts spacecraft: the Phase II capability provides a quantified, auditable confidence assessment of each vendor's digital twin against the authoritative reference, ahead of the on-orbit checkout window.

The second pathway is commercial transition through RTSync's existing customer base and product line. RTSync's prior Phase II transitions in mission-engineering and digital test and evaluation supply established pathways into Missile Defense Agency programs, Air Force Research Laboratory programs, and other Department of Defense customers, with the Bayesian tranche-evolution engine and the morphism quality framework as differentiating capabilities. The Phase II software, ontology artifacts, and methodology are packaged for RTSync's standard product delivery model.

The third pathway is research-to-transition through the University of Arizona, including continuing peer-reviewed publication of the underlying mathematics, transition of the formal foundation through Small Business Innovation Research Phase III follow-on awards or non-Small Business Innovation Research governmental funding sources identified in section 9.b, and educational integration through the Systems and Industrial Engineering Department's research and teaching programs. Phase II Task 1's draft journal manuscript provides the academic anchor for this pathway; the team's recent independent assessment exercise and the team's recent Mission Thread Ontology publication establish the methodology's operational credibility for follow-on funding agencies.

The team specifically anticipates Phase III opportunities at the intersection of mission engineering and continuous learning, with the Bayesian tranche-evolution engine as the mathematically rigorous core. The proposed capability is directly applicable beyond PWSA: any system-of-systems undergoing rapid iteration with multi-vendor performer ecosystems and quantitative readiness requirements is a candidate domain. Adjacent Department of Defense applications include other proliferated constellations, distributed counter-air-defense networks, and large-scale autonomous systems testing. Adjacent civil applications include critical infrastructure assurance and certification-driven domains such as commercial aviation.

(11) Key Personnel

Dr. Bernard P. Zeigler, Chief Scientist, RTSync Corp. (Principal Investigator), Citizenship – USA
Ph.D. Computer/Communications Sciences, University of Michigan, MI, 1968. M.S. Electrical Engineering, M.I.T., Boston MA, 1964. B.S. Engineering Physics, McGill University, Montreal CA, 1962.

Dr. Bernard P. Zeigler is Chief Scientist at RTSync and Professor Emeritus of Electrical and Computer Engineering at the University of Arizona. Dr. Zeigler, having published numerous papers and books on modeling and simulation, is best known for his theoretical work concerning modeling and simulation performed according to general systems theory, DEVS. Dr. Zeigler is a Fellow of the Institute of Electrical and Electronics Engineers (IEEE) as well as The Society for Modeling and Simulation International (SCS) which he served as President (2002-2004). He is a member of the SCS Hall of Fame. In 2013 he received the Institute for Operations Research and Management Sciences (INFORMS) Simulation Society Distinguished Service Award. In 2015 he received the INFORMS Lifetime Professional Achievement Award, which is the highest honor given by the Institute for Operations Research and the Management Sciences' Simulation Society. In 2022, Dr. Zeigler was given the ACM SIGSIM Distinguished Contribution Award. Dr. Zeigler led the development of the DEVS-based methodologies—System Entity Structure (SES), Experimental Frames (EF), and the hierarchy of system specifications—that form the mathematical substrate for ontology-driven model families, morphisms, and automated V&V, all central to the Phase II technical approach.
His recent work on model homomorphisms, SES abstraction, and DEVS-MBSE integration directly supports Phase II objectives including:

- the formal substrate for tranche-evolution reasoning,
- the ingestion and composability framework,
- automated KPP reverse-flow and V&V traceability, and
- the Bayesian morphism-based confidence engine.

Dr. Zeigler's leadership ensures continuity from Phase I, reduces technical risk, and anchors the project in the state-of-the-art of formal modeling and mission-engineering science.
Publications most relevant to the current application;

1. Zeigler, Bernard P., Alexandre Muzy, and Ernesto Kofman. *Theory of Modeling and Simulation*. 3rd ed. Academic Press, Elsevier, 2018.
2. Zeigler, B. P., Simulation-based Evaluation of Morphisms for Model Library Organization, in Model Engineering for Simulation, Eds: Lin Zhang, Y. Luli, and B. P. Zeigler, Model Engineering for Simulation, Elsevier 2019

3. Zeigler, Bernard P. 2022. “DEVS and MBSE: A Review.” *International Journal of Modeling, Simulation, and Scientific Computing* 13 (2). https://doi.org/10.1142/S179396232240002X.
4. Zeigler, Bernard P. 2023. “Extending the Hierarchy of System Specifications and Morphisms with SES Abstraction.” *Information* 14 (1): 22. https://doi.org/10.3390/info14010022.
5. Zeigler, Bernard P., C. Koertje, and C. Zanni. 2024. “The Utility of Homomorphism Concepts in Simulation: Building Families of Models from Base-Lumped Model Pairs.” *SIMULATION* 100 (12): 1181–96. https://doi.org/10.1177/00375497231199892.
6. Zeigler, B. P. Discrete Event Systems Theory for Fast Stochastic Simulation via Tree Expansion. Systems 2024, 12, 80. https://doi.org/10.3390/systems12030080

Dr. Doohwan Kim, President and CTO, RTSync Corp. (Senior System Engineer), Citizenship – USA

Ph.D. Computer Engineering, University of Arizona, Tucson, AZ, December 1996, M.S. Computer Engr, University of Arizona, 1994, B. Engr., Computer Engineering, Chonbuk National University, South Korea 1991

Dr. Doohwan Kim is a senior systems engineer and the President of RTSync Corp., with more than 25 years of experience leading the development, transition, and commercialization of DEVS-based modeling and simulation technologies. His work spans MBSE integration, autonomy and AI collaboration systems, digital engineering, high-performance computing, data engineering, and agent-based test instrumentation for DoD and federal agencies. Dr. Kim has served as Principal Investigator or senior technical lead on multiple large-scale R&D programs involving mission engineering, digital test and evaluation (dT&E), and system-of-systems simulation, making him uniquely qualified to guide the DEVS-PWSA Phase II effort. His leadership ensures continuity across RTSync’s DEVS technology stack—including ParaDEVS, CAMLI, DVAST, and DEVS-based Digital Twin - each of which directly supports the Phase II objectives of tranche-evolution reasoning, multi-performer model ingestion, and automated V&V. Dr. Kim also serves on the U.S. Technical Advisory Group (TAG) to the International Organization for Standardization (ISO), contributing to global standards in modeling, simulation, and systems engineering. His involvement in ISO/TC activities strengthens the project’s alignment with emerging digital-engineering standards and best practices.

Publications most relevant to the current application;

1. Zeigler, BP, Kim, D., Keller, N., and Ceney, J., Supporting the Reuse of Algorithmic Simulation Models, Summer Sim, July 2020 San Diego, CA, USA
2. Zeigler, B. P. and Kim, D., Multi-resolution modeling for adaptive UAV service systems, Summer Sim Conf. 2019. DOI: 10.23919/SpringSim.2019.8732861
3. Seo, C., Zeigler, B. P., and Kim, D., Expanding DEVS and SES Applicability: Using M&S Kernels within IT Systems, Symposium on Theory of Modeling & Simulation – DEVS (TMS/DEVS) Spring Sim 2014 San Diego, CA, USA (Best Theory paper for SpringSim Multiconference, 2014)
4. Alshareef, A and Kim, D., Handling complexity of large high-level specifications in simulation environment: the case of transformational model-battle management, The Journal of Defense Modeling and Simulation: Applications, Methodology, Technology, December 2024 https://doi.org/10.1177/15485129241305428

Mr. Christian Koertje, Senior Research Staff Member, RTSync, Citizenship – USA

M.S. Systems Science, Binghamton University (SUNY Binghamton), Binghamton, NY (2023),
B.S. Physics and B.A. Applied Mathematics, University of North Carolina Wilmington, Wilmington, NC (2021)

Mr. Christian Koertje is a senior research staff member at RTSync and one of the primary developers advancing the next generation of DEVS-based mission-engineering technologies. His work spans high-performance parallel and distributed simulation, DEVS-based MBSE integration, AI/ML-assisted model

generation, and model-family coordination, all of which are central to the DEVS-PWSA Phase II technical objectives. Mr. Koertje played a key technical role in both SDA Phase I efforts. He also contributed to the first DVAST Phase II project by designing the DEVS-based simulation-step architecture and developing a test harness for V&V, experience that directly supports Phase II's automated KPP reverse-flow and V&V traceability.

Publications most relevant to the current application;

1. Zeigler, B., Koertje, C., Zanni, C., Yoon, S., & Dutan, G. (2024, December). Advanced tutorial on paratemporal simulation using tree expansion. In 2024 Winter Simulation Conference (WSC) (pp. 1-15). IEEE.
2. Zeigler, B. P., Koertje, C., & Zanni, C. (2024). The utility of homomorphism concepts in simulation: building families of models from base-lumped model pairs. SIMULATION, 100(12), 1181-1196.

Dr. Paul Wach, Researcher, The University of Arizona (PI, Arizona), Citizenship – USA

Ph.D. Industrial & Systems Engineering, Virginia Tech, Blacksburg, VA, 2022. M.S. Mechanical Engineering, University of South Carolina, Columbia, SC. B.S. Biomedical Engineering, Georgia Tech, Atlanta, GA, 2009.

Dr. Wach leads the University of Arizona contribution to this Phase II effort as Researcher in the Department of Systems and Industrial Engineering. His research line, formal systems-theoretic foundations for mission engineering, model morphisms, and Bayesian test and evaluation, directly underwrites the Phase II DEVS-PWSA objectives: the formal substrate for tranche-evolution reasoning (a systems-theoretic ontology family and a library of isomorphic patterns), the morphism-quality metrics of structural and behavioral distance behind mission-thread comparison, the evidence-typed Bayesian tranche-evolution engine, and automated key-performance-parameter reverse-flow with verification-and-validation traceability. As principal investigator of record for the Phase I effort, he carries the technical line intact to the University of Arizona, ensuring continuity from Phase I and a low-risk institutional transition.

Publications most relevant to the current application;

1. Wach, P., Kamien, D., and Mantravadi, C. (2026) Authoring Mission Threads with Semantic Technologies: From Narrative Workflows to Executable, Explainable Knowledge. Semantic Technologies for Intelligence, Defense, and Security (STIDS) Conference, Arlington, VA.
2. Wach, P., Philipbar, B., and Gregory, J., Kim, D. (2026) Math-Based Data Structures and Analysis for Mission Engineering. INCOSE International Symposium 2026, Yokohama, Japan 2026.
3. Philipbar, B., Wach, P., and Kim, D. (2026) DEVS-Based Agentic AI Swarms: System Entity Structure of Queens for Rapid Systems Development and Verification. INCOSE International Symposium 2026, Yokohama, Japan 2026.
4. Wach, P., Sandman, K., and Iyer, A. (2026) Toward a Library of Isomorphic Patterns for Systems Engineering. Conference on Systems Engineering Research 2026.
5. Wach, P., Krometis, J., Sonanis, A., Verma, D., Panchal, J., Freeman, L., and Beling, P. (2022) Pairing Bayesian Methods and Systems Theory to Enable Test and Evaluation of Learning-Based Systems. INSIGHT, 25(4), 65-70.

Dr. Alejandro Salado, Associate Professor & Director, Systems Engineering, The University of Arizona, Citizenship – USA

Ph.D. Systems Engineering, Stevens Institute of Technology, Hoboken, NJ, 2014. MS Electronics Engineering, Polytechnic University of Catalonia, Spain, 2010. MS Project Management, Polytechnic University of Catalonia, 2009. MEng Space Systems Engineering, Technical University of Delft, The Netherlands, 2008. BS/MS Electrical and Computer Engineering, Polytechnic University of Valencia, Spain, 2007.

Dr. Salado, Associate Professor and Director of Systems Engineering at the University of Arizona, contributes the verification-and-validation and evidence-reasoning foundations central to Phase II. His work on verification-strategy design, set-based definition of verification activities, and the Bayesian formation of belief from verification evidence (National Science Foundation CAREER award) directly informs the Phase II evidence-type ontology (established, inferred, synthetic) and the Bayesian tranche-evolution engine, while his ontology-based digital Test and Evaluation Master Plan (dTEMP) research supports the Phase II automated verification-and-validation traceability.

Publications most relevant to the current application;

1. P. Wach, B. Zeigler, A. Salado, "Conjoining Wymore's Systems Theoretic Framework and the DEVS Modeling Formalism: Toward Scientific Foundations for MBSE," Applied Sciences 11: 4936, 2021.
2. J. Gregory, A. Salado, "An Ontology-based Digital Test and Evaluation Master Plan (dTEMP) Compliant with DoD Policy," Systems Engineering 27: 1012-1026, 2024.
3. J. Gregory, B. Jackson, A. Salado, "Automated Generation of Bayesian Verification Strategies using Semantic Web Technologies," Conference on Systems Engineering Research (CSER), 2025.
4. A. Kulkarni, A. Girod, P. Xu, A. Salado, "Optimal Verification Strategy for General Design Plans Using a Belief-based Approach," Systems Engineering 27: 74-98, 2024.
   H. Fiore, P. Pandit, J. Gregory, A. Salado, "An Ontological Representation of Mission Engineering Patterns for Digital Engineering," INCOSE International Symposium, 2026.

(12) Facilities/Equipment

The RTSync Team will support the proposed effort with RTSync and the subcontractor owned facilities and equipment. The RTSync Team anticipates no purchases of equipment to complete the proposed effort. The RTSync Team anticipates no requirement for Government Furnished Property to accomplish the proposed effort.

RTSync operates two strategically located offices in Chandler, Arizona and Oakland, California, enabling close collaboration with clients and facilitating on-site coordination to ensure alignment with customer requirements. These locations serve as hubs for project management, technical development, and client engagement. Each RTSync facility is fully equipped with essential infrastructure, including high-speed internet access, telecommunication systems for seamless connectivity, and dedicated workspaces designed to support both individual and collaborative tasks, and standard utilities to maintain a productive and comfortable working environment. The primary office in Chandler, AZ accommodates up to 10 employees and benefits from proximity to major academic institutions. RTSync leverages technical resources and expertise from University of Arizona (Tucson, AZ) to enhance its research and development capabilities. RTSync holds a Secret-level Facility Clearance (in process), enabling seamless transition of DEVS-PWSA capabilities into classified environments for SDA, MDA, and Space Force programs.

Regarding technical equipment and development resources, ss a simulation software development company, RTSync maintains a robust suite of hardware and software tools to support high-performance computing and accurate simulation modeling through commercial cloud computing services. RTSync utilizes its proprietary MS4 Me Cloud Toolset, a powerful platform built on the DEVS framework, to support the design and implementation of both general systems and Systems of Systems (SoS). This cloud-based environment enables integration of system structure and component behavior into a unified modeling and simulation platform. To support the development of DEVS-PWSA technology, RTSync employs cloud services from Microsoft Azure, Google Cloud, and Amazon AppStream, utilizing virtual machines.

The facilities where the proposed work will be performed meet environmental laws and regulations of federal, state, and local Governments for, but not limited to, airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid and bulk waste disposal practices, and handling and storage of toxic and hazardous materials.

The UA will support this project with its Digital Engineering Factory (DEF). The DEF is a digital engineering environment that hosts multiple engineering tools, integrated in a 'hub-and-spoke' arrangement with the Violet tool at the 'hub'. The DEF has been implemented on premises. A cluster of 320 CPUs has been configured to provide approximately 100TB of usable storage for installations and to host work. The distribution has been configured to support high-availability load balancing and ease of maintenance. Users access the tools available on the DEF via their web browser, where they can log in to web applications (e.g., Jama, Jira) via their UA credentials or virtually run software that has been installed on the server (e.g., Cameo Systems Modeler, Genesys). While the DEF has not been deployed in classified environments yet, it has been designed to be deployed into such environments if necessary. In addition, the UA has SCIFs and a robust export compliance (e.g., ITAR, EAR), with active Department of State registration, and industrial security program to cover all levels of restricted information from Controlled Unclassified Information (CUI) through Top Secret and special access programs for both DoD and Intelligence Community agencies.

(13) Consultants/Subcontractors

RTSync partners with The University of Arizona (UA) under this STTR Phase II effort, with UA performing approximately 31 percent of the proposed work as a subaward. The UA contribution is led by Dr. Paul Wach, Researcher in the Department of Systems and Industrial Engineering, who served as principal investigator of record on the SF24B-T009 and SF24C-T003 Phase I efforts and provides direct technical continuity from Phase I into Phase II. Dr. Alejandro Salado, Associate Professor and Director of Systems Engineering at UA, contributes complementary expertise in verification-strategy design, ontology-based digital Test and Evaluation Master Plan formulation, and MBSE co-pilot research, drawing on his NSF CAREER program and a portfolio of recent Stevens Institute of Technology Systems Engineering Research Center (SERC) Web-Based Research Tasks (WRTs) in digital engineering and test-and-evaluation transformation. The UA subaward also provides access to the Department's Digital Engineering Factory (DEF), a 320-CPU on-premises digital-engineering hub, and to UA's Department of State export-compliance program (International Traffic in Arms Regulations and Export Administration Regulations), Sensitive Compartmented Information Facilities (SCIFs), and Top-Secret-cleared industrial-security program. Together, RTSync and UA jointly address harmonization of systems engineering, systems theory, digital engineering, mission engineering, advanced data analytics, and the decision complexity inherent in tranche-based PWSA evolution.

Complementing Dr. Wach's leadership, Dr. Alejandro Salado, Associate Professor and Director of Systems Engineering at UA, contributes deep expertise in verification-strategy design, ontology-based digital Test and Evaluation Master Plan (dTEMP) formulation, and MBSE co-pilot research. Dr. Salado's contributions draw on his NSF CAREER award and a portfolio of Systems Engineering Research Center (SERC) Web-Based Research Tasks (WRTs) focused on digital-engineering transformation, model-based V&V, and mission-thread alignment. His work strengthens the Phase II approach to automated KPP reverse-flow, V&V traceability, and the integration of formal methods with MBSE artifacts. Together, Dr. Wach and Dr. Salado provide a rare combination of systems theory, systems engineering, and digital-engineering expertise, directly aligned with SDA's need for a rigorous, scalable, and auditable mission-engineering environment.

The UA subaward also provides access to critical institutional infrastructure that materially enhances the Phase II effort. This includes the Digital Engineering Factory (DEF)—a 320-CPU on-premises digital-engineering hub that will host the integrated DEVS-PWSA prototype and support scenario execution, ontology

maintenance, and multi-performer model ingestion. UA also provides access to its export-compliance program (ITAR/EAR), Sensitive Compartmented Information Facilities (SCIFs), and a Top-Secret-cleared industrial-security program, ensuring that the project can securely handle controlled technical data and coordinate with SDA's secure digital-engineering ecosystem. These facilities enable the team to work with high-fidelity models, sensitive mission-thread data, and evolving tranche-specific artifacts in a compliant and secure environment.

Together, RTSync and UA form a unified team capable of addressing the full spectrum of challenges inherent in tranche-based PWSA evolution. RTSync contributes the DEVS-based simulation engine, ParaDEVS stochastic extensions, ingestion-pipeline architecture, and agentic-AI orchestration capabilities. UA contributes the systems-engineering rigor, ontology-driven digital-engineering methods, and secure computational infrastructure needed to integrate these components into a coherent mission-engineering environment. This partnership harmonizes systems engineering, systems theory, digital engineering, mission engineering, advanced data analytics, and decision-support methodologies, ensuring that the Phase II effort delivers a transition-ready capability aligned with rigorous assessment at the pace of tranche evolution.

(14) Prior, Current, or Pending Support of Similar Proposals or Awards

No prior, current, or pending support for proposed work.

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